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REVIEW

Dye-sensitized solar cells: A promising solution for climate change

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Abstract

Climate change is a growing global issue that affects all organisms, environmental elements, and biodiversity on Earth. This significant challenge underscores the urgent need for worldwide collaboration to address and overcome it. In response, governments, businesses, and the global economy are making substantial efforts to mitigate the negative effects of climate change on the environment. There are numerous obstacles in the fight against climate change, including the need for decarbonization, the protection of water resources and green spaces, and the implementation of innovative solutions across the energy, transportation, and construction sectors. Renewable energy, particularly solar energy generated through solar cells (SCs), is widely regarded as a promising strategy for tackling climate change and reducing its adverse effects. This approach has proven to be the most effective in addressing the current environmental crises.

Highlights

- Dye-sensitized solar cells (DSSCs) provide a low-cost and eco-friendly PV alternative.
- Advances in dyes, photoanodes, and electrolytes improved efficiency and durability.
- DSSCs exhibit a low carbon footprint.
- Limitations of Solar Cell Technology.
- Hybrid Integration of solar cells with other Renewable Energy Systems.
- Combined renewable technologies can ensure grid stability and continuous power supply.
- Solar-driven decarbonization strategies reduce fossil fuel dependency and emissions.

Keywords Renewable energy · Photovoltaic · Dye sensitized solar cell · Climate changes

Introduction

The global population is growing exponentially, influenced by technological advancements. Fossil fuels still dominate the energy sector despite being unsustainable and having significant negative impacts on the environment and human health [1, 2]. The burning of fuel releases Considerable amounts of greenhouse gases into the atmosphere. These gases include nitrous oxide, carbon dioxide, and methane. With the rapid pace of civil and industrial development, it is predictable that emissions of these gases will increase over time. Climate change is regarded as a serious health issue, with rising sea levels, ecosystem disruption, and other challenges expected due to current and predicted levels of greenhouse gases, assuming no changes to energy sources [3]. Humans face risks from climate change and health issues related to fossil fuels [4, 5]. In light of these facts, each country has begun implementing various strategies to prevent catastrophic situations. To reduce these issues, governments have begun reevaluating their energy plans and regulations. Developing effective strategies can lead to a reduction in greenhouse gas emissions and address related issues in a comprehensive protocol. Different recommended approaches include creating new, efficient gadgets with less environmental impact [6, 7] and shifting partially or fully to renewable energy sources [8, 9]. In the near future, the most practical way to phase out fossil fuels is to rely on renewable energy resources (RERs). Renewable energy sources can deliver sustainable energy services by utilising readily available

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materials and local resources that are cost-effective due to their environmental abundance [10–12]. Sunlight offers humanity a more affordable, sustainable, and clean energy source. The estimates for global energy consumption history from 1990 to 2040 provided credence to these ideas. This estimate indicates that between 2090 and 2100, solar energy will account for more than 60% of all energy [13]. Figure 1 shows the Common categories of renewable energy sources.

Annual Energy Outlook 2022 (AEO2022) Reference scenario, which considers the existing laws and regulations, projects that the percentage of renewable energy generated in the United States will increase from 21% in 2021 to 44% by 2050. For Egypt, the supply of electricity generated from renewable sources increased from 20% in 2022 to 42% in 2035 (Fig. 2). Wind and solar energy make up the majority of this rise in renewable energy. By 2050, the proportion of hydropower in the energy mix is going to remain unchanged, while the combined share of other renewable energy sources, such as geothermal and biomass is expected to be less than 3%.

The gradual expansion of natural gas-fired generation and the ongoing retirement of coal plants would cause the overall percentage of fossil fuel-fired power generation in the United States to drop from 60% to 44% in the AEO2022 Reference, as predicted in the meantime. The overall share of natural gas in the generation mix decreases from 37% in 2021 to 34% in 2050, although the absolute output from natural gas-fired sources increases [14]. The governmental initiatives and market forces promote renewable energy sources, while generation from these sources increases to offset the declining percentages of coal and nuclear power.

Solar energy (Photovoltaics PV)

There has been significant interest in photovoltaic (PV) technology, which utilizes the PV effect to convert sunlight into electricity. Solar radiation is an abundant energy source for the Earth, providing approximately 1.2 × 10^14 kW nearly 10,000 times the current global consumption. In fact, the energy the Earth receives from the sun in just one hour is equivalent to a year's worth of energy

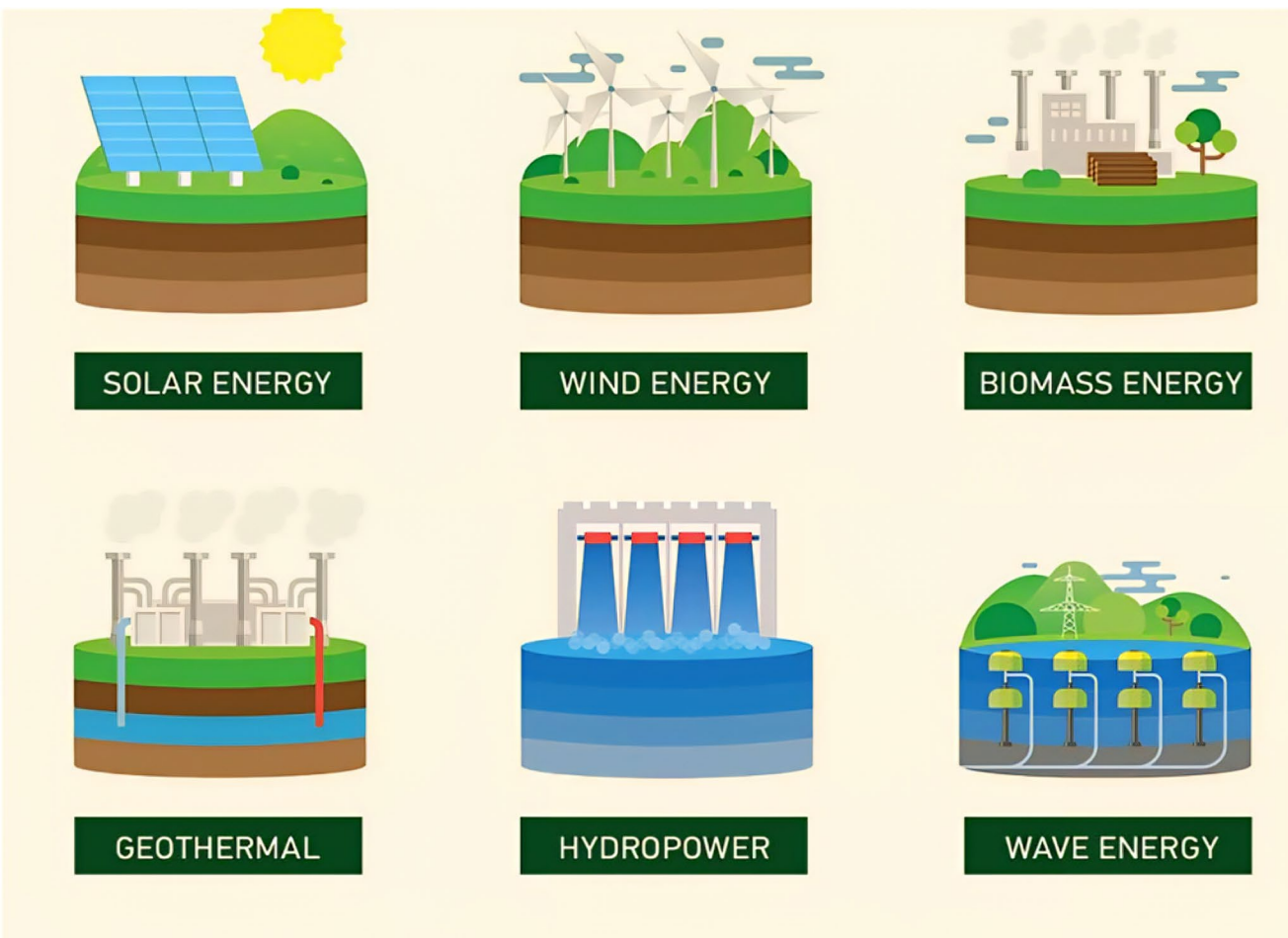

Fig. 1 Renewable energy sources

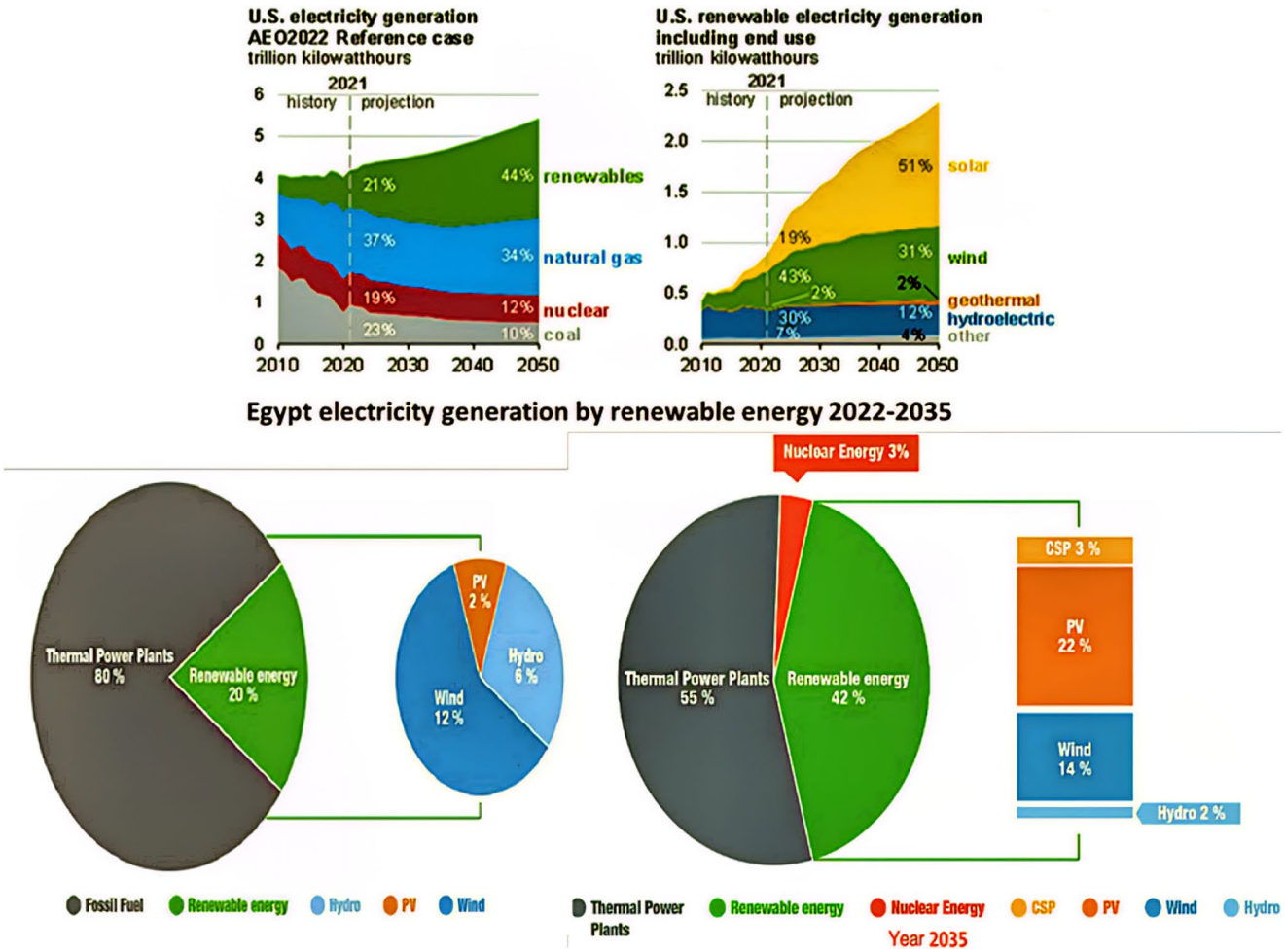

Fig. 2 Generation of electricity from renewable sources

consumed by humanity [15]. The PV effect arises from the potential difference between two semiconductor layers: one layer is present between the semiconductor and the metal. In contrast, the other is due to the opposite conductivity. PV device operates by generating electrons and holes through photo-generation within a light-absorbing material. These charge carriers are then separated and transmitted via conductive contacts, which allow electrical current to flow between the semiconductor layers [16]. PV cells can be categorized based on their composition, operational mode, and manufacturing methods. There are three generations of solar cells currently in development: first, second, and third generation [17]. First-generation solar cells are made from either monocrystalline or polycrystalline silicon. However, their adoption has been limited due to high manufacturing costs and complex processing requirements [18, 19]. Consequently, second-generation solar cells gained more popularity. Regarding the third generation of SCs, they include the lately uncontrollably spreading dye-sensitized SCs (DSSCs) [20–29], Organic tandem SCs (OTSCs) [30, 31], Inorganic SCs (INSCs) [32], perovskite SCs (PSCs) [33, 34], and quantum dot SCs (QDSCs) [35–40].

Dye-sensitized solar cells (DSSCs) and perovskite solar cells (PSCs) demonstrate strong performance in low-light conditions, making them ideal for Internet of Things (IoT) and indoor applications. In contrast, silicon solar cells (SiCs) generate insufficient energy in similar low-light scenarios. Additionally, DSSCs and PSCs possess transparent designs, which make them suitable for Building-Integrated Photovoltaic (BIPV) applications, such as smart windows. On the other hand, SiCs are opaque. When it comes to stability, SiCs have superior long-term stability and durability compared to DSSCs and PSCs. The latter two types of cells face ongoing challenges related to stability under harsh conditions, which researchers are actively working to improve. However, DSSCs and PSCs are cost-effective, making them a more accessible choice for various applications. In contrast, SiCs can pose a barrier for certain BIPV applications due to their higher costs (Table 1).

Table 1 Shows comparisons of different SCs in terms of construction, efficiency, and stability

Types Of SCs	Mapping Component	Materials	PCE%	Stability	Ref
SiCs	Two primary areas: performance mapping (using characteristic performance maps to analyze efficiency and thermal behaviour) and physical mapping (imaging surface potential or silicon composition to identify defects and optimize design).	**phosphorus-doped n-type Monocrystalline silicon wafers**	**27.03**	**25 years**	[41, 42]
DSSCs	physical arrangement and interconnectivity of its four main functional components	**Tio_2 BPHA/SL9/[Cu+/2+(tmby)2](TFSI)1/2 (tmby, 4,4′,6,6′-tetramethyl-2,2′-bipyridine PEDOT DSSC**	**15.2%**	**90 days**	[43, 44]
PSCs	These three components form the basic structural unit of a perovskite, identifying and analyzing its components (A, B, and X sites) using various techniques	**FTO/PCBM/$Cs_{0.05}(MA_{0.17}FA_{0.83})_{0.95}Pb(I_{0.83}Br_{0.17})_3$/MeO-2PACz/Au PSC**	**34.75**	**One year**	[45, 46]

The global use of PV has grown between 1992 and 2022. It changed during this time from being a specialized market for small-scale applications to becoming a common source of electricity. Its capacity and production have grown at a rate of about 26% each year between 2016 and 2022, roughly doubling every three years. Several countries introduced feed-in tariffs and other subsidy schemes to encourage investment in solar PV systems, which were then seen as a viable renewable energy source. Consequently, experience curve effects, economies of scale, and technological advancements led to a significant drop in the cost of solar.

The increase in PV usage was facilitated by a number of national efforts, including China's 2011 five-year plan for energy production, the United States' Million Solar Roofs Initiative, and Germany's Energiewende [47]. Since then, the global deployment of PV has accelerated and is now more competitive with traditional energy sources. Utility-scale plant sales began to take off in the early 21 st century as a supplement to rooftop and other distributed applications. Nearly 30 nations achieved grid parity by 2015. Following the introduction of commercial solar cell manufacturing in the 1950 s, several countries have emerged as the global leaders in solar PV electricity production. The United States was the first, followed by Japan, Germany, and now China. Up from roughly 3% in 2019 [48], the total installed PV capacity worldwide by the end of 2022 was 1,185 gigawatts (GW), providing more than 6% of the world's electricity needs. In 2022, for nine countries, solar PV accounted for more than 10% of the total annual household electricity consumption; Spain, Greece, and Chile accounted for more than 17%. Bloomberg New Energy Finance (BNEF) predicts that global solar installations will increase, in 2019, an additional 125–141 GW, for a total capacity of 637–653 GW by year's end. The International Energy Agency's (IEA's) high-renewable scenario predicts that by 2050, solar PV capacity will reach 4.7 terawatts (4,674 GW), with more than half of this capacity located in China and India. This will make solar power the world's largest source of electricity [49].

The size of the Egyptian Solar (PV) Market is projected to grow from 2,300 MW in 2023 to approximately 3,546.96 MW by 2028, achieving a compound annual growth rate (CAGR) of 9.05% during the forecast period (2023–2028). Key factors contributing to this growth include the declining prices of solar PV modules, supportive government policies, and increasing energy demand. As part of its Vision 2030 plan, Egypt aims to elevate the share of renewable energy to around 30% by 2050 and reduce rising carbon emissions. Additionally, the country has an untapped solar PV generation capacity of 74,000 TWh annually. This scenario is expected to create significant opportunities for market growth in the coming years.

Several factors directly influence the output of solar power plants, including site design, the ability to connect to the electricity grid, the orientation of the panels in relation to the sun's rays, and hourly fluctuations in energy prices. An important aspect to consider is the analysis of

environmental factors, which can be categorized using the DEEPLIST framework (Demographic, Economic, Environmental, Political, Legal, Informational, Social, and Technological) [50]. For PV farms, topography plays a crucial role. The suitability of the terrain for installing PV panels significantly affects energy generation efficiency. Elements such as shading, slope, and location can all impact the overall performance of a PV system [51] (Fig. 3).

The combination of solar cells with wind or geothermal energy systems on the environment and climate was discussed. Despite coordinated attempts to advance and invent energy production technologies separately, there are currently few studies that consider the environmental sustainability of integrated renewable power plants. A promising development in energy innovation is the integration of diverse renewable resources into a coherent power generation system [52]. Suggested an approach: Integrating several renewable energy sources into a single structure to improve the energy grid's efficacy, flexibility, and dependability (guaranteeing a steady and dependable power supply). The environmental effects of three integrated renewable-based power plants are evaluated. Solar, wind, and geothermal resources are available. Three case studies have been of interest for consideration: combined geothermal-wind (CGW), combined solar-geothermal (CSG), and combined solar-wind (CSW) cycles. There are many excellent benefits to these coupled systems. The solar-wind combination had the least influence on ozone depletion, and the combined geothermal-wind combination had the least impact on freshwater and marine ecotoxicity. According to the study, CO_2 emissions are reduced by 28% for the combined solar-geothermal and 56% for the combined solar-wind scenarios when the lifespan of perovskite solar cells is extended from three to fifteen years. Among all assessed scenarios, the combined solar-wind, combined solar-geothermal, and combined geothermal-wind options are the most sustainable in terms of ozone depletion, freshwater ecotoxicity, marine ecotoxicity, and climate change impacts, respectively. For optimal environmental sustainability, this research recommends investing in the optimal combination of integrated power cycles utilizing both established and emerging renewable technologies.

Limitations of solar cell technology

Efficiency constraints Solar (PV) technologies still have inherent and practical efficiency limitations despite tremendous technological advancements. Under typical test conditions, the Shockley–Queisser limit, the thermodynamic

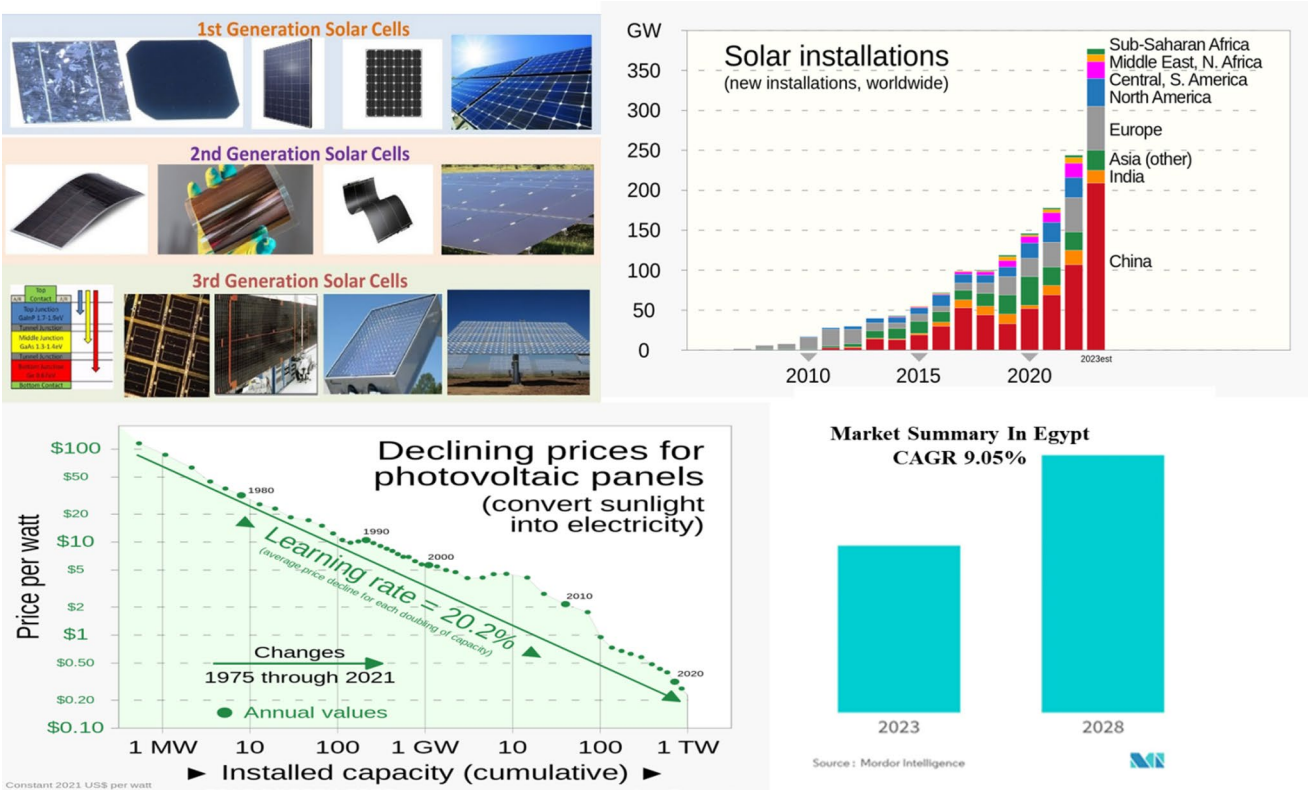

Fig. 3 Types of SCs and PV cell production

upper limit for a single-junction cell, is approximately 33.7% [53]. Due to losses from charge-carrier recombination, imperfect light absorption, thermalization, and spectral mismatch between the solar spectrum and semiconductor bandgap, commercial silicon modules typically achieve an efficiency of 15–22% in real-world operation [54]. Furthermore, performance is further deteriorated by unfavourable working conditions such as high temperatures and low irradiance. Higher theoretical ceilings are provided by multi-junction and tandem architectures, but practical deployment at high efficiency is hindered by their complexity, cost, and scaling hurdles [55].

Intermittency and energy storage Due to daily cycles, seasonal changes, and weather fluctuations, solar energy production has an intermittent nature. As a result, photovoltaic power systems alone cannot guarantee a stable electricity supply. The overall energy structure becomes more expensive and complex due to this intermittency, necessitating backup or supplementary storage solutions to ensure reliability. Moreover, matching variable solar energy inputs with the demand of the electrical grid remains a major obstacle to the large-scale integration of PV power [55].

Material, manufacturing, and cost limitations Over the past decade, PV module costs have decreased; however, manufacturing and material limitations have persisted. Concerns regarding resource availability and environmental safety are raised by the fact that thin-film technologies rely on rare or hazardous materials, such as indium, gallium, and cadmium, and high-purity silicon fabrication remains energy-intensive [54, 56]. Furthermore, complex passivation and heterojunction processes in advanced cell architectures can raise production costs or create yield issues at scale [54].

Issues with lifespan and degradation PV modules gradually deteriorate in use due to dust buildup, humidity, thermal cycling, and UV exposure. Depending on technology and environmental factors, annual degradation rates might vary from 0.2% to 3%. It has been demonstrated that rust and dust alone can lower efficiency by up to 1.9% and 3%, respectively. The long-term energy yield and economic feasibility are directly impacted by these degradation mechanisms, underscoring the need for reliable encapsulation materials and designs.

End-of-Life management and recycling challenges End-of-life (EoL) management is becoming a crucial sustainability concern as PV capacity grows globally. Module heterogeneity, connected layers, and the presence of dangerous substances like lead and cadmium hamper recycling procedures. Although several recycling methods have been proposed, they remain unstandardized and are not supported by regulations, nor are they commercially viable [57, 58]. For the PV industry to grow sustainably, strong circular economy methods must be established.

Hybrid integration of solar cells with other renewable energy systems

The potential of hybrid systems, which combine photovoltaic (PV) solar cells with other renewable energy sources such as wind, biomass, hydroelectric, and incorporate energy storage technologies, to enhance energy sustainability, efficiency, and reliability, has become more widely acknowledged. The advantage of these arrangements is the availability of complementary resources; for example, wind energy production frequently peaks at night or during cloudy conditions when solar output is low, resulting in a more consistent and balanced power supply [59, 60]. Depending on the location and climate, studies show that solar-wind hybrid systems can increase overall energy consumption by 20–40% when compared to standalone PV systems [60]. In a similar vein, hybrid solar-hydro and solar-biomass systems have shown improvements in capacity factor and system reliability of 15–25% [61]. In addition to improving grid stability and lowering reliance on fossil fuel backup, the combined use of these resources can reduce greenhouse gas emissions from local energy systems by up to 30% [61]. To fully reap the benefits of hybrid renewable systems, however, obstacles such as system complexity, control optimisation, and higher capital costs still need to be overcome; efficient energy management techniques and storage integration are essential [62].

A strong route towards the deep decarbonization of energy systems is provided by hybrid energy systems (HES), which combine conventional backup generation, energy storage (batteries, hydrogen), and renewable-generating technologies (such as solar PV and wind turbines). According to modelled data, a hybrid PV, wind, and battery system in Zagazig, Egypt, reduced CO_2 emissions by about 33.9% when compared to the baseline grid-only scenario [63]. The integration of variable renewable energy sources (VRE), usually solar photovoltaic (PV) and wind, with energy storage (such as batteries) and occasionally backup or dispatchable generation, is a common modelling assumption in the design of hybrid energy systems (HES) for deep decarbonisation. For instance [63], used meteorological data and load profiles to size a hybrid design of PV panels, wind turbines, converters, and battery storage in a case study for Zagazig, Egypt. A net CO_2 reduction of 156.355 tonnes yr^{-1} and the lowest net-present cost (US$1.361 billion) among the tested scenarios were achieved by the optimal solution, which included 181.000 kW of PV feeding directly into the grid

[64]. Solar irradiation (4.6–5.8 kWh/kWp/d), wind resource density (~ 650 W/m2), battery round-trip efficiency (~ 85%), and a 20-year system lifetime were among the key assumptions. According to this analysis, the scenario's suggested renewable proportion (energy from PV + wind) was roughly 35–45% of yearly generation, with corresponding carbon savings of about 30–40% when compared to a baseline based only on fossil fuels. The impact of hybridization and storage architecture on system performance measures including capacity factor, capacity credit, and flexibility is examined in a second technical step of modelling. The authors of a study of utility-scale PV-wind-battery systems in Texas simulated hourly dispatch under different wind-to-PV and battery-to-PV capacity ratios and discovered that, in some configurations, PV-wind-battery hybrids achieved capacity factors of 60–80%+ and capacity credits that approached 100% of point-of-interconnection (POI) capacity [65]. The modelling included curtailment affects hourly market prices (energy + capacity) as a "price-taker," lithium-ion battery storage with a 4-hour discharge time, and battery round-trip efficiency of about 90%. According to their findings, bigger battery capacity in fewer complementary systems may perform similarly to smaller battery capacity (e.g., battery: PV ratio ~ 0.25) in a well-matched PV-wind location. In high-complementarity sites, for example, a battery-to-PV capacity increase from 0.25 to 1.0 only enhanced the capacity factor by ~ 10–15% points, indicating decreasing returns [65]. This analysis highlights the significance of appropriate size and resource complementarity. These findings suggest that it is possible to achieve CO_2 emission reductions of more than 50% in comparison to conventional generation by designing hybrid systems that push renewable-generation shares above 50–70% of annual production while retaining high dependability.

Modelling assumptions and system design The performance envelope and consequent decarbonization potential are usually defined by a variety of baseline assumptions in the modelling of hybrid energy systems (HES). First, depending on resource quality, storage availability, and backup generation capacity, it is frequently expected that renewable energy sources (RES), such as solar-PV and wind, will provide a primary generating fraction, typically in the range of 30% to 70% of total annual energy output. For instance, some studies estimate a wind resource with a mean power density of around 500–800 W/m² and solar irradiation of approximately 4.5–6 kWh/kWp/d, so that the combined RES capacity objective is set to meet approximately 40–50% of the yearly load in an off-grid or weak grid HES [66]. The usual model for storage is a bank of lithium-ion batteries with the following specifications: a round-trip efficiency ranging from 85% to 90%, a depth-of-discharge limit of approximately 80%, a lifetime round-cycle deterioration of about 0.8% per year, and an expected design life of 10–15 years [67]. Backup generation, which is typically a diesel or gas turbine, is designed to address the residual load if renewable energy sources (RES) and storage is insufficient. Its utilisation is frequently restricted to ensure that fossil backup generation does not exceed 20–30% of annual generation in a viable decarbonisation scenario. Lifetime horizon modelling is typically conducted over a period of 20–25 years, with discount rates ranging from 6% to 10% and an annual escalation of fuel costs of approximately 2% − 3% [67]. The levelized cost of energy (LCOE) model, which is used in many HES studies, accounts for both capital expenditure (CAPEX) and operational expenditure (OPEX) for all components, including backup fuel, storage, converters, RES, and balance of system. As a result of continuous technology advancements, it is commonly assumed that CAPEX for solar-PV will decrease at a rate of around 5%–8% annually and battery storage at a rate of about 7%–10% annually [67]. RES curtailment is frequently allowed under the operational plan to prioritise reliability over maximum renewable yield (e.g., 5%–10% of yearly RES output). Hourly or sub-hourly time-series (e.g., 8,760 h/year) are employed in dispatch models, and the optimisation may seek to maximise the renewable portion subject to cost limitations or minimise the overall system cost subject to dependability (e.g., > 99.9% availability) [68]. As an illustration of the diminishing returns of storage sizing, in a sample PV-wind-battery HES, raising the battery-to-PV capacity ratio from 0.25 to 1.0 increased the renewable fraction from approximately 45% to approximately 60%, but only improved reliability or capacity credit by approximately 10–15% points in high-complementarity resource locations [69].

Performance metrics and decarbonization outcomes Renewable energy fraction (REF, % of annual generation from RES), fossil backup fraction (FBF, % of annual generation from conventional fuel), hybrid system capacity factor, capacity credit of RES plus storage, system reliability (e.g., loss-of-load probability), and CO_2 emissions reduction relative to a baseline fossil scenario are important performance metrics in HES modelling. A typical modelling result, for example, might show a REF of 50%, FBF of 25%, with storage/residual imports covering the remaining amount, and a CO_2 emission reduction of about 40%–55% in comparison to a diesel-only baseline [70].

When storage and RES are designed optimally and dispatch is effectively handled, some advanced scenarios in large-scale grid-connected HES modelling show REF values of 60%–80% and corresponding CO_2 reductions of

~ 60%–70% [71]. State-of-charge (SoC) range (usually 20–90%), curtailment rate, and battery cycling frequency (e.g., 250–300 full cycles/year) are additional metrics. To evaluate the robustness of decarbonisation and cost outcomes, sensitivity analyses are frequently carried out to modify the RES resource quality (± 20% irradiance/wind), storage cost (± 30%), fuel price escalation (± 1%–4%), and discount rate (4%–12%). Pareto-frontiers showing trade-offs between cost and emissions are frequently produced by multi-objective optimisation; for instance, a 10% increase in storage may only lower emissions by 3–5%, capturing diminishing marginal returns [68]. Therefore, if storage is sized appropriately and resources are complementary, modelling indicates those HES can realistically achieve one-third to two-thirds emissions reductions and renewable shares up to ~ 70% with current technology cost assumptions.

Implications for review and outlook The decarbonization potential of HES is significantly influenced by resource complementarity (e.g., solar + wind), storage efficiency and cost trends, system reliability restrictions, and the operational/dispatch approach, as these modelling assumptions and indicators demonstrate. Although optimisation frameworks have advanced, significant gaps remain in life-cycle carbon assessments, behavioural models, and dynamic grid integration, as the current review demonstrates [72]. To achieve a REF beyond 80 per cent and CO_2 reductions exceeding 70%, future models must incorporate demand-side flexibility, sector coupling (electricity, heat, hydrogen), and emerging storage technologies (e.g., hydrogen, flow batteries) into consideration (Table 2).

DSSCs

(DSSCs) are being extensively researched as a clean and sustainable energy source. Compared to other types of solar devices, DSSC technology is considered more promising. Its mode of operation significantly differs from that of traditional solar cells, as it emphasizes interfacial processes rather than the bulk processes that govern silicon p-n junctions and other conventional cells. DSSCs consist of four main components: the counter electrode (CE), working electrode (WE), redox mediator (electrolyte), and sensitizer (dye). In

Table 2 Typical modelling assumptions in hybrid energy system (HES) decarbonization studies

Parameter	Typical Range/Assumption	Description/Notes	References
Solar irradiance (GHI)	4.0–6.5 kWh/m²/day	Annual mean global horizontal irradiation assumed for PV yield modelling.	[66, 68, 70]
Wind power density	400–800 W/m² (at 50 m hub height)	Typical onshore sites; affects sizing of wind capacity in hybrid configuration.	[66, 68]
Renewable energy fraction (REF)	30–70%	Fraction of annual generation supplied by RES (PV + wind + others).	[66, 67, 70]
Battery round-trip efficiency	85–92%	Lithium-ion baseline; includes inverter losses.	[67, 69]
Depth-of-discharge (DoD)	70–90%	To balance usable capacity and cycle life.	[69]
Battery lifetime	10–15 years (≈ 4 000–6 000 cycles)	Used for cost and degradation modelling.	[69, 70]
Converter/inverter efficiency	93–96%	Accounts for DC–AC/DC conversion losses.	[69]
Backup generator share	≤ 20–30% of annual generation	Used to maintain reliability during low-resource periods.	[66, 68]
System reliability target	≥ 99.9% availability	Often expressed as loss-of-load probability < 0.1%.	[68]
Discount rate	6–10%	Used in LCOE and net-present-cost (NPC) calculations.	[66]
Fuel price escalation	2–3% per year	Reflects expected fossil-fuel cost trends.	[68]
CAPEX reduction trend	PV – 5%–8% yr^{-1}; battery – 7%–10% yr^{-1}	Anticipated technology learning rates.	[67, 70]
Battery : PV capacity ratio	0.25–1.0	Higher ratios increase renewable penetration but show diminishing returns beyond ≈ 0.5.	[69]
Curtailment allowance	5–10% of RES output	Applied to manage grid stability and oversupply.	[68, 70]
Simulation horizon	20–25 years (8 760 h per year)	Typical for techno-economic optimisation models.	[68]
CO_2 reduction potential	35–70% vs. fossil baseline	Range achieved by optimised PV-wind-storage HES.	[66, 70, 71]

1991, Professor Grätzel and his colleagues first developed DSSCs [73]. The global market for DSSCs was valued at USD 90.5 million in 2019 and is projected to grow at a compound annual growth rate (CAGR) of 12.4% from 2020 to 2027 [74]. The greatest application sector for DSSCs is in portable charging, where demand is increasing, and the market is expected to meet this need effectively (Fig. 4).

Device structures of DSSCs

The sandwich cell, which is the typical DSSCs device structure, has conducting glass substrates for both the WE and CE positioned face-to-face, with a thin layer of the redox electrolyte sandwiched in between Fig. 5. Usually measuring around 25 mm, the thermoplastic frame that serves as the sealing determines the spacing between the electrodes. Even smaller spacing is preferable since it reduces resistance brought on by the electrolyte's redox mediator diffusion. Glass coated with fluorine-doped tin oxide (FTO) is the most often used conducting glass in DSSCs. High solar light transmittance, low sheet resistance, and strong chemical and thermal stability are all well-compromises with FTO glass. The mesoporous TiO_2 layer is sintered on top of FTO glass, which serves as the photoelectrode. Between the FTO and the mesoporous TiO_2, there is an optional thin and dense TiO_2 layer known as the "blocking layer," whose purpose

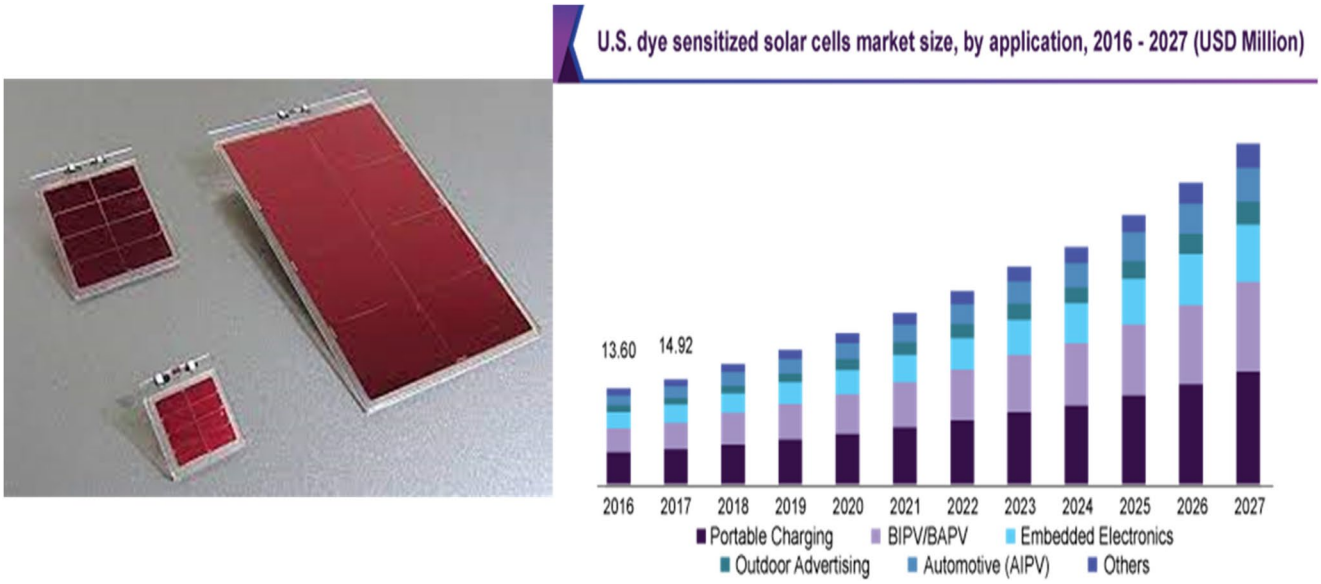

Fig. 4 Global DSSCs Market Size

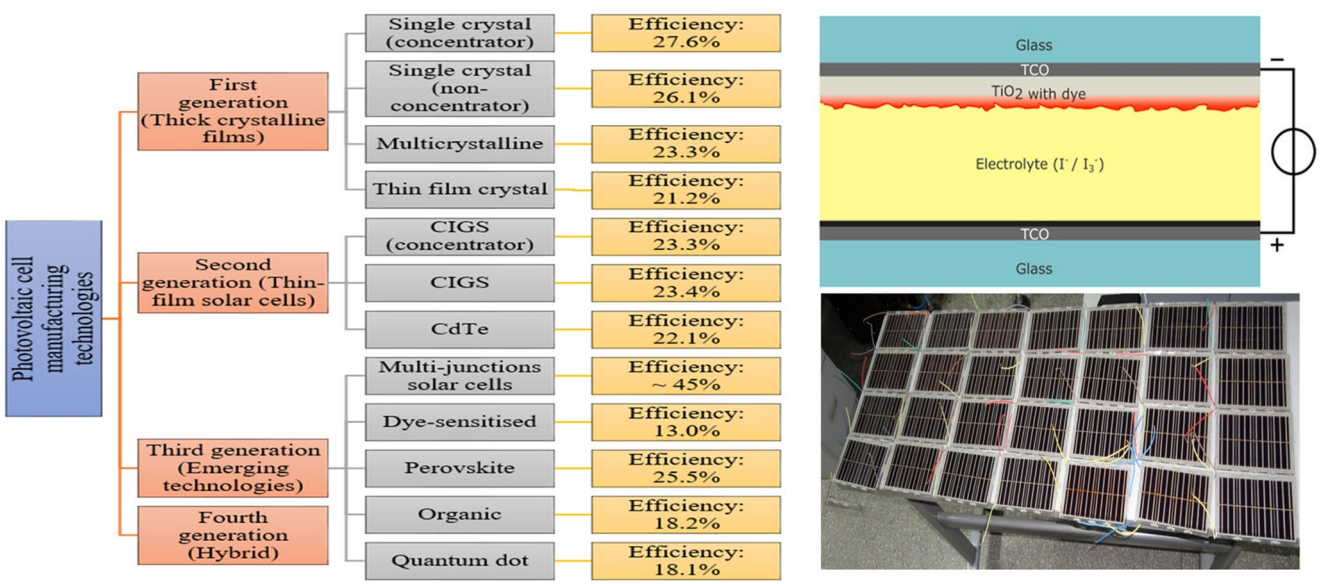

Fig. 5 Fabrication of DSSCs

is to reduce electron recombination from the FTO to the redox electrolyte. To enhance light capture in the device, a light-scattering TiO_2 layer can be placed on top of the mesoporous layer. The catalyst, which could be Platinum (Pt), carbon, or a conducting polymer, is coated onto FTO glass to create the CE. Also Fig. 5 showed a diagram representing efficiencies of different types of SCs. Some commonly used active materials in each type of SCs are also mentioned. The efficiencies of DSSCs devices based on the new materials are yet to match that of silicon SCs. Researchers across the globe are still optimistic due to the rapid progress in the parameters of these higher-generation solar cells, in addition to low material production cost of such SCs.

Understanding the behaviour of electrons and the intricate workings of the DSSCs is crucial for improving the performance of every component of the system. This kind of SCs mechanism is a replication of the steps in a plant's photosynthesis cycle [75]. As seen in Fig. 6, the dye molecule is raised from its ground state to its excited state by light absorption. As a result, electrons are moved into the semiconductor's conduction band. Donor electrons found in the iodide electrolyte convert the oxidized dye back to its ground state, enabling the oxidized dye to absorb photons [76]. The Pt on the CE catalyzes the electrons accepted by the electrolyte, which recombine with triiodide to form diiodide. This process leads to the conversion of sunlight to electricity [77]. This mechanism was illustrated in Eqs. (1–7).

$$\text{Anode}: \ h\upsilon + O \rightarrow O^* \tag{1}$$

where hυ is the incoming photon and O* is the excited anthocyanin dye. The excited dye will inject electrons into the conduction band of TiO_2 according to the equation shown in (2).

$$O^* \rightarrow O^+ + e^-{}_{(cb)}\,(TiO_2)\ \text{Injection Process} \tag{2}$$

The excited dye will be oxidized. The oxidized dye molecules will be reduced by an Iodide ion, which in turn becomes a tri-iodide ion, following the equations given by:

$$2\,O^+ + 3I^- \rightarrow I^-{}_3 + 2O\text{Regeneration} \tag{3}$$

$$\text{Cathode}: I^-{}_3 + 2e^- \rightarrow 3I^- \tag{4}$$

$$O^* + e^-{}_{(cb)}\,(TiO_2) \rightarrow O^+ + 2e^-{}_{(cb)}\,(TiO_2) \tag{5}$$

$$I^-{}_3 + 2e^-{}_{(cb)}\,(TiO_2) \rightarrow 3I^-{}_3 + (TiO_2) \tag{6}$$

$$\text{Cell}: 2e^-\,(CE) + h\upsilon \rightarrow 2e^-\,(TiO_2) \tag{7}$$

Two crucial factors of PV performance are the fill factor (FF) and power conversion efficiency (PCE). Equations (8) & (9) are used in their calculation. The three primary processes involved light absorption, generation of electric charges, and transport of electrons and holes are facilitated by three distinct materials. Because of this feature, (DSSCs) are highly interfacial devices, which distinguishes them from other types of solar cells based on their mechanism. SCs can generate their maximum voltage, or open-circuit

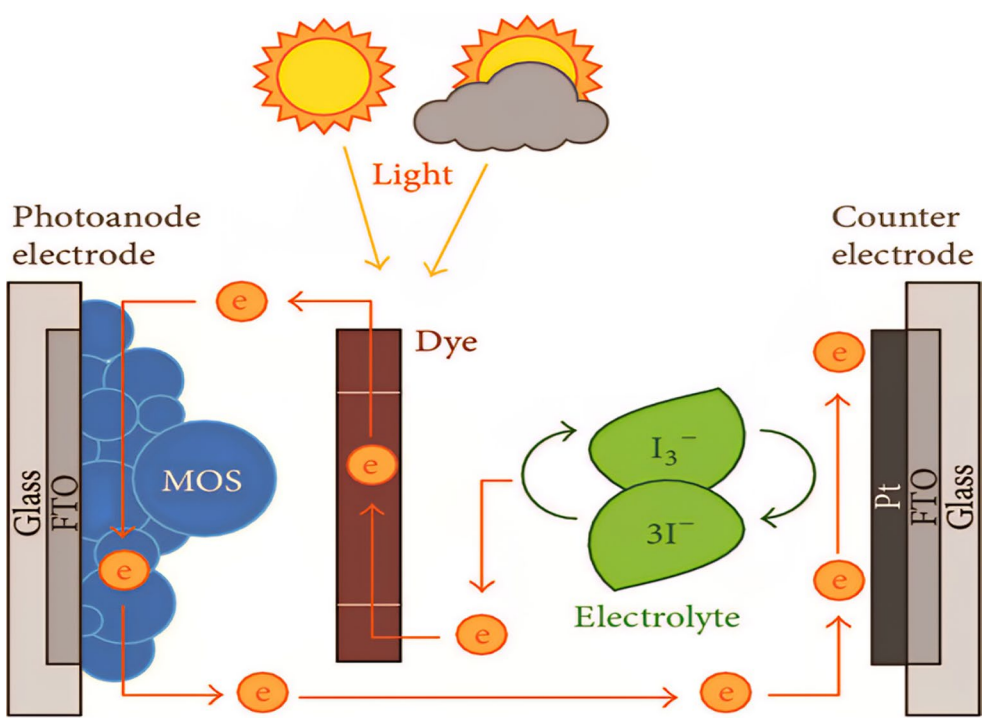

Fig. 6 Mechanism of the generation of electrons in dye sensitized solar cell

voltage (V_{oc}), when no current flows. Due to the bias on the SCs' junction when light-generated current flows through it, Voc indicates how forward-biased the SCs are. Short-circuit current (I_{SC}) refers to the current flowing through an SC when there is no voltage across it, often called a short circuit. (FF) is calculated by dividing the maximum power (P_{max}) by Voc and I_{SC}. The quality of SCS is assessed using a parameter called FF. The P_{max} output from an SC is determined by the FF parameter, along with V_{oc} and I_{SC}.

The power can be assessed through comparing the (P_{max}) with the theoretical power generated from (V_{oc}) and (I_{sc}). (FF) can be determined using Eq. 8:

$$FF=(V_{max} \times J_{max})/(V_{oc} \times J_{sc}) \tag{8}$$

In this equation, Vmax represents the maximum voltage, and Jmax represents the maximum current. (PCE) can be assessed by dividing the energy output by the energy input. PCE indicates the fraction of incident power that is extremely converted into electrical energy. It can be expressed as:

$$\begin{aligned} PCE\,(\%) &= (V_{max} \times J_{max}/P_{in}) \times 100 \\ &= (V_{oc} \times J_{sc} \times FF/P_{in}) \times 100 \end{aligned} \tag{9}$$

In these calculations, an input power of 100 mW/cm² or 1000 W/m² is commonly used for efficiency assessments. The (SCs) efficiency is estimated under AM1.5 conditions at 25˚C. The temperature of the cell, along with the spectrum and intensity of the incident sunlight, affects the PCE. Therefore, it is essential to carefully control the parameters used for measuring efficiency to enable accurate comparison of one device's performance against another [78].

Recent advances in photo-anode

A key component of the DSSCs is the photoanode (PA), which acts as the overall energy conversion centre. The PA is made from dye-sensitized semiconductor nanoparticles deposited over a transparent conducting glass. For the photo-anode in DSSCs, a variety of semiconductor oxides have been employed, including TiO_2, ZnO, SnO_2, and WO_3 [79]. TiO_2 is the most widely used semiconductor in DSSCs due to its abundance, affordability, biocompatibility for wearable applications, non-toxicity, and excellent crystallinity and dispersion. Of the three stable polymorphs of TiO_2 (Brookite, Rutile, and Anatase), the brookite phase is the least popular due to the complex method needed in its production. Although both forms perform similarly, anatase is preferred over rutile in solar energy conversion due to its improved charge transport, higher specific area, and higher electron Fermi level. In addition, the anatase bandgap is 3.2 eV while the rutile bandgap is 3.0 eV, meaning that the anatase form has greater conduction-band energies than the rutile form. Sol-gel synthesis, hydrothermal synthesis and solvothermal synthesis [80] are some of the methods that are used to create TiO_2 nanostructures. Compact TiO_2 coatings for electron-blocking materials were produced using different techniques, including electrochemical deposition [81], thermal oxidation [82], spray pyrolysis [83], spin coating [84], and pulsed laser deposition [85] (Fig. 7).

Over the years of development, several modifications to TiO_2 have been made to improve the PCE of DSSCs. Morphology [58], Doping [86], and Composites-hybrids [87] are the three categories of modifications to TiO_2.

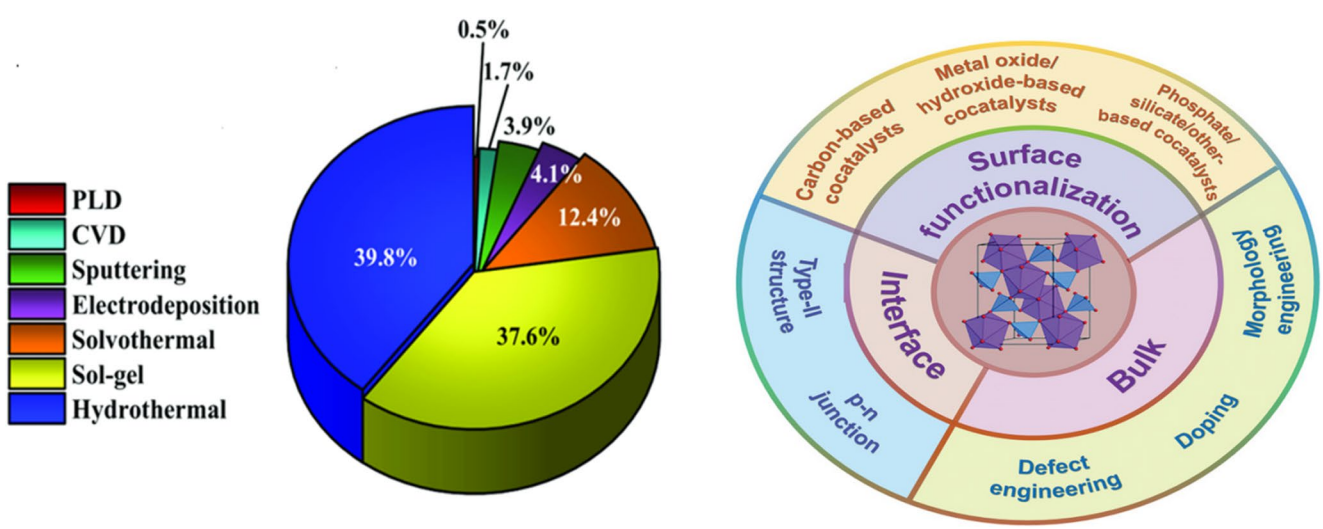

Fig. 7 Different synthesis and advances of photoanode

Morphology Increasing the total amount of dye adsorbed on TiO_2 is the goal of optimizing the morphology. As the quantity of sensitizing molecules on a semiconductor material's surface grows, so does its light-harvesting efficiency (LHE). The adsorption of the dye is enhanced by improving the dye's anchoring on the semiconductor's surface or by expanding the semiconductor's surface area. Numerous research endeavours have examined the variations in cell performance resulting from modifications to the porosity and particle size of TiO_2 Fig. 8. Zhang et al. [88] created mesoporous TiO_2 microspheres for use as the PA in DSSCs using a single-step hydrothermal process. The TiO_2 microsphere's structure enhances charge collection efficiency, reduces charge recombination, prolongs exciton lifespan, and produces good light scattering. The PCE of the double/layered DSSCs with TiO_2 microspheres as a scattering layer is 5.61% compared to that of the single P25 layer (2.67%) and the single TiO_2 microspheres layer (2.14%).

Khakpour et al. [95] produced a (3D/TiO_2 nanorod (3D/TiO_2/NR) as WE, and they reported a rise in efficiency over the control. They evaluated the PV properties of I_2-based and Co (II/III) tris(2,2′-bipyridine) complex electrolytes that were built with two distinct WE, namely (3D/TiO_2/NR) and TiO_2-Nanoparticle (TNP). The results showed that utilizing bulky electron shuttles in 3D-TiO_2-NR-based PA greatly enhances the PV properties of DSSCs.

Mo et al. [96] created TiO_2 microspheres (MS/TiO_2) for a WE. The synthesized MS/TiO_2 exhibits excellent light scattering properties and a superior dye loading capacity. As a result, the gadget displayed a high efficiency of 10.35%.

Zan et al. [97] created hierarchical flower-like TiO_2 microspheres (FMS) with an exceptional ability to scatter light and absorb dyes. FMS reported to have an average size of 15 nm and a high specific surface area of 118.6 $m^2\ g^{-1}$.

Hierarchical TiO_2 nanotubes (HTNT) were manufactured by Chen et al. [98]. The prepared HTNTs are constructed from one-dimensional (1D) long TiO_2 nanotube trunks and many short TiO_2 nanorod branches. In comparison to P-25 (8.34%), the gadget produced an efficiency of 9.89%. Another TiO_2 was prepared with several morphologies [99–102] in an effort to boost the efficiency of DSSCs.

Doping The intentional replacement of semiconductor atoms with impurities in the host material—typically using an ion of similar atomic radius to the ion being substituted in the lattice—is known as substitutional doping. This process can modify the electronic band structure of semiconductors. Research conducted by Dahlan et al. [103] demonstrated the effects of copper (Cu) doping in two-dimensional titanium dioxide nano-walls (Cu/TNW). They found that doping TiO_2 nano-walls with Cu enhances their carrier transport and photoactivity characteristics. Additionally, this doping reduces charge recombination in (DSSCs) and lowers the charge transfer resistance at the interface between the

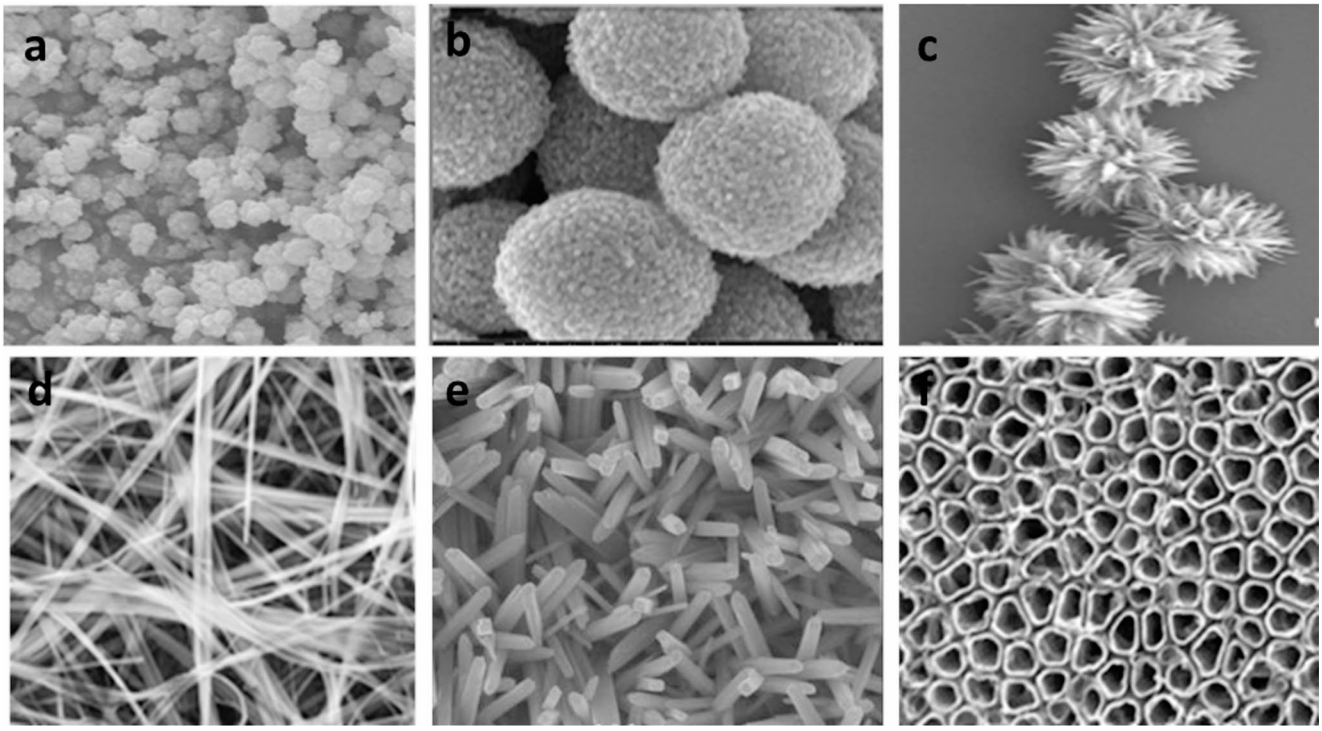

Fig. 8 Different nanostructures of TiO_2 (Derivated from refs [89–94])

electrolyte and the semiconductor. These improvements are essential for increasing power conversion efficiency. The efficiency of the S/TiO_2 nanofiber-based anode, created by Mahmoud et al. [104], was 4.27%, while the efficiency of the undoped anode was 1.54%. Dimethyl sulfoxide is hydrothermally treated for 18 h, which helps dope TiO_2 nanofibers with sulfur atoms. The presence of titania and the sulfur 3p state permits a decrease in the bandgap and an increase in the valence bandwidth, resulting in an increase in the PCE of DSSCs. Prabavathy et al. [105] successfully created Ca/TiO_2 nanorod structures utilizing a hydrothermal one-step procedure. They used an I_2-based electrolyte, Pt CE, and a rose sensitizer on WE to create DSSCs. According to optical studies, the band difference of the TiO_2 sheet reduced from 3.08 eV to 2.75 eV as Ca doping increased. It was observed that less TiO_2 adheres to FTO substrates when the proportion of Ca doping increases. Burak et al. [106] conducted a study on doping titanium dioxide (TiO_2) with copper (Cu) and manganese (Mn) for use as a working electrode (WE) in (DSSCs). They reported that the power conversion efficiency (PCE) of the DSSCs was 1% for the doped TiO_2 compared to 0.5% for the undoped anode. The doping process altered recombination and charge transfer dynamics. Furthermore, doping with ions that can interact with the sensitizer and bind these dyes to the working electrode could potentially enhance the PCE of the DSSCs. Bramhankar et al. [107] developed a TiO_2 PA co-doped with nickel (Ni) and zinc (Zn). Additional PA prepared with various ions [108–110] in an effort to improve the PCE of DSSCs.

Composites-hybrids Composites and hybrid materials improve the PCE of DSSCs because the higher surface area supplied by the composite structure allows more dye molecules to adsorbed, increasing the quantity of light that collected. A composite of high conductivity materials, such as carbon derivatives, can improve the extraction and mobility of photo-generated electrons, resulting in higher photocurrents [111]. Additionally, a synergistic combination of different materials can result in higher PV performance, including increased I_{SC} and V_{oc}. Composites designed for enhanced light scattering and a broad absorption spectrum, enabling the collection of more photons across the solar spectrum. To minimize charge recombination losses on PA surfaces, facilitate more electrons to reach the external circuit. By combining different materials, we can engineer features such as optical bandgap energy to better align with the requirements of dye sensitizers and the solar spectrum. Additionally, enhanced synergistic interactions among the components of a composite can lead to improved stability for the PA material.

The conduction-band make up of titanium (3d orbitals) while valence-band (VB) composed of (hybridized/oxygen 2p) and titanium (3d orbitals) for TiO_2. Decreases in recombination processes and the quantity of electrons returned to the VB are the results of this. Composites made from one or more components can be utilized in (DSSCs) and can take the form of layered frameworks or core-shell structures. The properties of TiO_2 composites can be modified by altering the material composition, which can change the bandgap or reduce the rate at which charge carriers recombine. A double/layered WE composed of (TiO_2 NRs) over layer and (TiO_2 NPs)/ZnO NF) was prepared by Chen et al. [112]. This WE exhibits improved in (light scattering, charge transfer, and absorption of N719 dye). When compared to DSSCs using pure TiO_2-P25 nanoparticles, the device's efficiency was 8.01%, resulting in a 28.9% increase in PCE.

A hybrid WE based on/graphene nanoribbons (GNR) was described by Akilimali et al. [113], the PCE showed a 20% increase in comparison to the control device. Better dye loading, an extended electron lifetime, and a decrease in carrier recombination credited with this accomplishment. Depending on the ribbon diameter, GNRs offer intriguing properties including high specific surface area and high mobility electrons. Because of these properties, GNRs for insertion into the anode material to boost DSSCs efficiency might be a better choice than other carbon allotropes. Bakr et al. [114] prepared a tin oxide TiO_2 composite nanofiber (SnO_2/TiO_2NF) as WE, which demonstrated a 40% higher PCE in comparison to TiO_2-based DSSCs. Compared to TiO_2, the synthesized CNFs had greater electrical conductivity, and compared to SnO_2, they had higher CB energy. CNFs boost the VOC and decrease the transport resistance, which raises the efficiency from 5.1% to 8%. According to Cuong et al. [115], the addition of reduced graphene oxide (r-GO) to TiO_2 improves electron transport and lowers electron recombination in TiO_2/r-GO hybrid WE based DSSCs. Additionally, they used r-Go, MWCNT, and Pt to create a highly conductive hybrid CE. PCE of different ratios of (TiO_2NR/NP) composite was examined by Chatterjee et al. [116]. The composite at a 1:1 ratio demonstrated the maximum efficiency of 8.61%, surpassing both the NR and the NP-based photo anode by 29% and 14%, respectively. Several composites have reported to be applied as WE in DSSCs to improve their PCE.

Latest developments in sensitizers

Figure 9 illustrates the straightforward structure of the dye, which serves as a photosensitizer. The donor and acceptor parts of the dye and the donor and acceptor parts of electrons are connected by a π bridge. The acceptor portion of the dye contains the functional groups that attach it to the TiO_2. To capture visible light and start the electrical current, a Sensitizer dye is necessary in DSSCs.

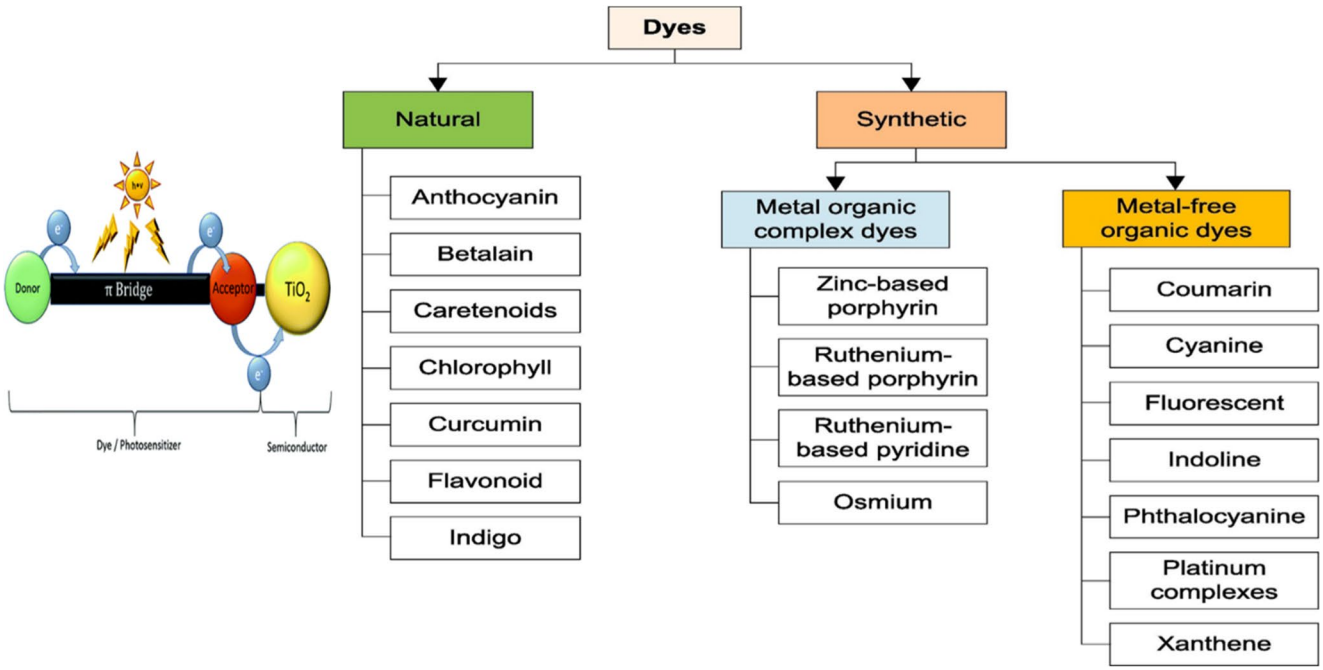

Fig. 9 Simple design and various type of dye

The chemical adsorption or adhesion of the natural dyes on the surface of the TiO_2 film occurs due to the presence of anchoring functional groups, such as (-OH) groups in the dye molecules, so it's a key component in capturing solar light energy on the photo anode. To create more effective dyes for DSSCs, essential design must be fulfilled. Before effective electron injection into the TiO_2 conduction band can occur, the sensitizing dyes must firmly bind to the surface of the photocatalyst (TiO_2). For the oxidized dye to regenerate efficiently, its highest occupied molecular orbital (HOMO) must be smaller than the hole transport material, and its lowest unoccupied molecular orbital (LUMO) must be higher than TiO_2's conduction band. Finally, to improve the effectiveness of the absorption process, the dye's light-harvesting capacity in the visible and/or near-IR regions should be significant [117], the cell's effectiveness is determined by the type of dye sensitizer (natural or synthetic).

(a) Natural dye; Natural resource dyes are another widely used DSSCs sensitizer. Recently, efforts have been made to develop ecologically acceptable DSSCs by extracting natural dyes from readily available sources [118]. Fruits, flowers, and leaves are few examples of naturally occurring materials that provide a wide range of colours and pigments that can be extracted and utilized in DSSCs [119, 120]. These natural dyes have several benefits, including abundant presence, easy manufacturing methods, high absorption coefficients in the visible range, and environmental friendliness [121, 122]. Due to their selective light absorption, natural dyes offer far less effectiveness than synthetic dyes. The highest efficiency was reported for Gambier Leaves, as a natural dye, has 4.71% [123].

(b) synthetic dyes; Synthetic dyes, particularly those based on metals like ruthenium complexes, have greater power conversion efficiency than natural dyes. They are frequently more stable and lasting under operational circumstances than natural dyes, which fade more quickly. These dyes are designed to exhibit excellent light absorption capabilities, consequently, facilitating efficient charge transfer to semiconductors. However, they confront many problems. The use of costly and rare noble metals, such as ruthenium, causes an increase in the cost of solar cells. Synthesizing these complex colours is frequently time-consuming and involves complicated chemical procedures. Many synthetic dyes and their precursors may contain harmful compounds, which can pose risks during production and present disposal issues at the end of their lifecycle. Synthetic dyes are divided into two main categories; Metal-organic complex (MOC) dyes and metal-free organic (MFO) dyes. Metal-free organic (MFO) dyes are organic compounds without metal atoms, while metal-organic complex (MOC) dyes include a metal ion with an organic ligand.

MCOs for practical use in DSSCs are their scarcity, purification challenges, environmental risks, and low molar extinction coefficients [124]. Due to their high molar extinction coefficients, ease of synthesis, low toxicity, great structural flexibility, and environmental friendliness, MFOs sensitizers have garnered more attention than their metal-based counterparts [125]. MFO sensitizer frequently contains an electron acceptor (A) in the form of a D-π-A, an electron donor (D), and a π-π-bridge [126]. Many chromophores, including carbazole, phenothiazine, and triphenylamine, serve as electron donors, each possessing unique electronic and steric properties [127–129]. π-bridges are integrated to enhance their light-harvesting capacity and facilitate intramolecular charge transfer. Various anchoring units are employed, such as [R − C(= O) − N(OH) − R', C5H5N, −OH, C6H6O2, $RCOO^-$, $C–PO(OR)_2$, RSO_3^-], among others successfully identified to bind the chromophore onto the semiconductor [130–132]. The maximum efficiency of the DSSC was achieved using SL10 as a synthetic dye, with a maximum PCE of 15.2%. They utilised a co-sensation strategy to apply hydroxamic acid pre-adsorption to develop the molecular packing of SL9. They found that the adsorption of hydroxamic acid onto the surface of titanium dioxide, before co-sensitization, facilitated the adsorption of dye molecules and prevented the absorption of sensitizing substances [43].

Recent advances in electrolyte

One of the most vital roles of the electrolyte in DSSCs is to accept electrons from the redox mediator, aiding the sensitizer in regenerating from its oxidized state. Charge transport between the WE and CE of the SCs, as well as the regeneration of the dye deposited on the semiconductor, is managed by the electrolytes in DSSCs [133]. For (PV) systems, the most critical factor is the chemical composition of the electrolyte. The electrolyte must be able to quickly reduce the oxidised dye and ensure a rapid flow of charge carriers between the porous nanocrystalline layer and the (CE) to facilitate efficient light energy conversion. Electrolytes can be categorized into three main types: liquid electrolytes (LEs), quasi-solid electrolytes (QSEs), and solid-state electrolytes (SSEs). A comprehensive overview of all the electrolytes developed for (DSSCs) is presented in Fig. 10.

LEs According to a publication [73], the first DSSCs were reported with an (LE) comprised of an organic solvent containing an iodide/triiodide redox pair with PCE ≈ 7%. This electrolyte solution was made without the use of any other

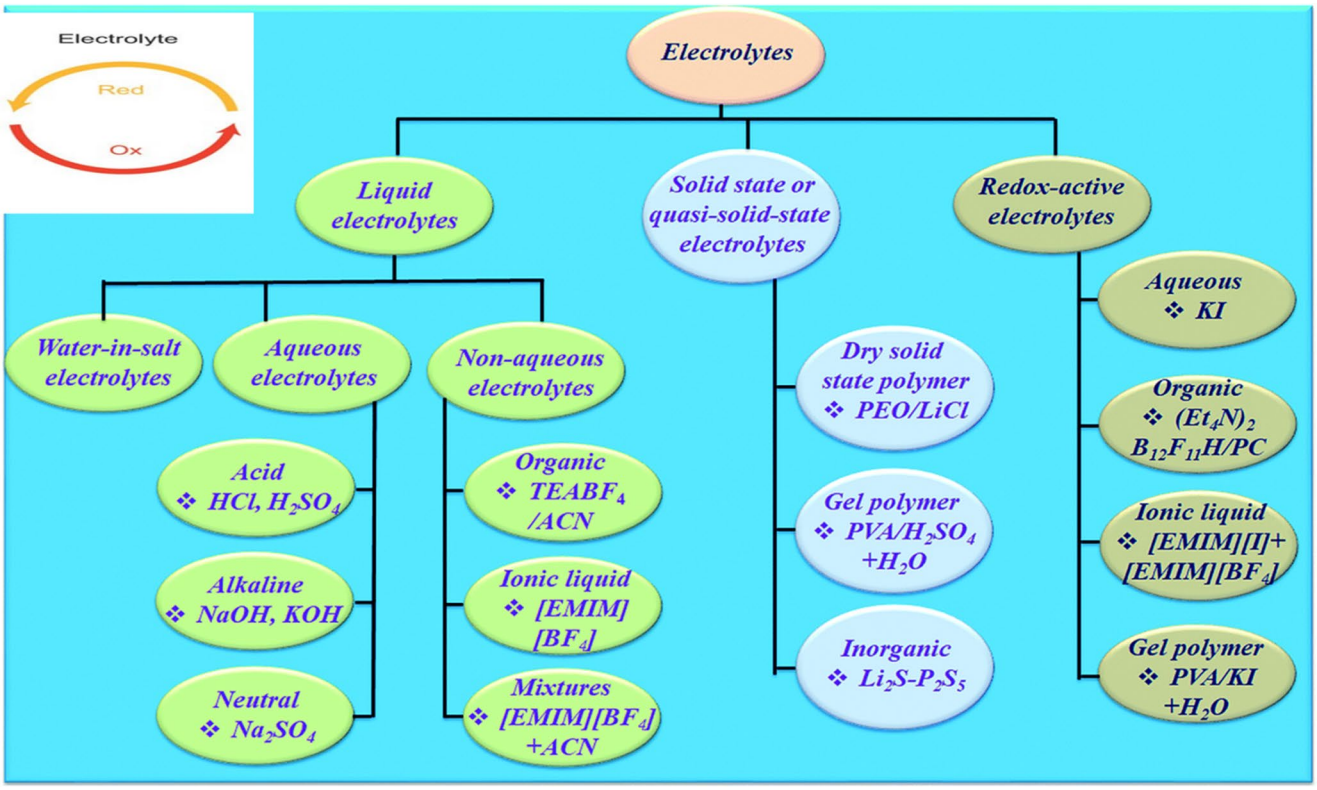

Fig. 10 Different types of electrolytes in DSSCs

additives. Subsequently, other electrolyte compositions using different solvents and extra additions were created. In addition to increased long-term PV performance stabilities, several of these have attained high PCE [134]. In comparison to other electrolytes, liquid solvent-based electrolytes have many benefits, including low viscosity, excellent conductivity, and good connectivity interaction at the electrode/electrolyte interface, and simple preparation techniques that result in high PCE [135]. It's interesting to note that, when exposed to full sunlight, DSSCs with LEs and conventional device architecture have the best PCE to date, which approaches 14% [136]. The Leakage problem from this cell, which reduces the initial PV performance, is the primary constraint for low viscosity LEs. To prepare different moderate viscosity solvents, solvents have been produced and have shown outstanding long-term performance stability [137]. Ionic liquids, iodide/triiodide redox shuttles, iodide/triiodide-free mediators, and organic solvents are among the LEs for DSSCs, as shown in Fig. 10. Iftikhar et al. [133] go into great length about these subclasses of LE.

QSEs Although DSSCs with LEs have the highest known solar-to-electrical conversion efficiencies, leakage and sealing problems that affect long-term device stability continue to be the principal obstacle to their successful commercialization [138]. Gel electrolytes as QSEs can be used in place of volatile LEs to get these restrictions [139]. A polymeric matrix serves as a framework for the solvent and inorganic salts added as additions in gel electrolytes, also known as QSE [140]. Either an inert or coordinating polymer matrix can be utilized to build a three-dimensional network [141]. The cations and anions in inert gel electrolytes, such as poly(vinylidene fluoride) (PVDF) and poly(acrylonitrile) (PAN), can travel freely in the solvent (liquid phase). In gel electrolytes containing coordinating polymers, such as poly (ethylene oxide) (PEO), ionic transport occurs in a binary phase that is both liquid and solid [142]. The gel system is seen as a hybrid structure that combines the cohesive and diffusive transport characteristics of a solid and a liquid, rather than being either one alone. Gel electrolytes are often simple to manufacture, with minimal solvent volatility, excellent chemical and mechanical stability, and good ionic conductivity (10^{-5}–10^{-3} S cm^{-1}). In addition to having a good contact between the CE and WE with excellent filling of the TiO_2 pores [143]. The high viscosity of gel electrolytes is another appealing feature that makes them suitable for roll-to-roll deposition techniques [144], which helps to lower the cost of manufacturing DSSC modules [145].

SSEs As discussed earlier, a common issue with employing LEs (liquid electrolytes) or QSEs (quasi-solid electrolytes) in DSSCs is their tendency to leak out of the cell channel, which can lead to degraded device performance, especially under stressful operating conditions [146]. This restriction encouraged many research teams to create fully solid-state DSSCs in which solid-state conducting materials (SSCMs) were used in place of the LE [146]. Because these SSCMs don't include any solvents, they have distinct advantages over the electrolytes covered in the preceding sections. This makes them particularly well-suited for large-area DSSCs modules. To replace LEs and QSEs with SSCMs, a wide range of materials has been researched and developed. In general, SSCMs include organic hole-transport materials, inorganic hole-transport materials, and ionic conductors [147, 148]. Additionally, the types of procedures used to introduce the electrolyte into the cell channel have a significant impact on the overall PV performance of DSSCs. There are two traditional techniques used to introduce electrolyte into laboratory-sized DSSCs: vacuum back filling and electrolyte injection filling [149]. First, holes are drilled in the side of the CE, and then the PE and CE are bonded to each other using thermoplastics to build the cell channel [150]. If a CE has one or two holes, one drop of electrolyte solution is placed on top of the hole and then inserted by establishing a vacuum [150]. If the CE has two holes, another method of introducing the electrolyte is the capillary action [151].

While traditional electrolyte-filling methods offer several benefits, they also present many drawbacks including the risk of glass electrodes shattering during mechanical drilling, increased costs due to the need for additional thermoplastic sealing, and the requirement for extra glass covers [152]. The uneven distribution of the electrolyte solution within the cell, along with variations in the performance of individual cell channels, is an additional issue caused by the molecular filtering effect produced by the mesoporous layers in DSSCs [150]. Restrictions may be lessened by using well-established printing and coating methods, such as screen printing or inkjet printing, for the electrolyte layer [152]. An electrolyte composition that is spatially homogeneous throughout the device can be achieved through screen or inkjet printing. However, the electrolyte viscosity must be optimized to use these techniques. By doing this, it prevents electrolyte leakage and improves the devices' long-term stability for large area [153] and lab-sized DSSCs [152].

Recent advances in CE

The two primary tasks of the CE, which are highly influential in DSSCs' efficiency overall, are (receiving electrons from the external circuit and transmitting them into the cell, with low resistance required) and acting as a catalyst for the reduction of the oxidized species of the redox mediator. Strong catalytic activity towards the redox mediator, high

conductivity, high reflectance, low cost, high surface area, high porosity, low charge-transfer resistance, high exchange current density, chemical resistance to corrosion, energy alignment matching the potential of the electrolyte's redox couple, and good processability for deposition are all desirable characteristics of a good CE for DSSCs [154]. There is a wide range of proven CE preparation procedures for DSSCs, including thermal and photo-decomposition [155], electrochemical deposition [156], chemical vapour deposition [157], and sputter deposition [158, 159]. The preparation techniques have a significant impact on the electrodes' catalytic and electrochemical properties as well as their surface, shape, and particle size. More catalytically active sites are provided by smaller particles and larger electrode surface areas, which also enable better electrode operation Fig. 11 [160]. Metal materials exhibit superior thermal stability, superior flexibility, and great electrical conductivity under high-temperature treatment. In addition to lowering costs, using metal materials for SCs electrode substrates can enhance cell performance by reducing the internal resistance.

CE including Pt Pt is first chosen as the CE material for DSSCs because of its excellent conductivity, catalytic activity for triiodide reduction, and high reflection performance [161]. A Pt layer of 1 nm thickness was sputtered by Cheng et al. [162] using thermal annealing at 400 °C on the FTO substrate. Next, the amorphous-nanocrystalline mixed Pt film became a highly crystalline one with improved transparency and electrocatalytic I_3^-reduction capability, suitable for use as CE of the double-sided DSSCs. The PCE of DSSCs was raised to 8.32% by Yang et al. [163] by combining silicon nanowires and Pt nanoparticles to create the CE. Firstly, it increased Pt activity by expanding its surface area. Second, silicon hydrogen bonds can inhibit the agglomeration of metal nanoparticles during catalysis by lowering the metal ions and encouraging the formation of small metal nanoparticles on the surface of silicon nanowires.

CE including non- Pt metal Despite the Pt electrode materials being undoubtedly excellent as CE. The Pt still faces several challenges, including the cost, scarce nature, inadequate stability, extended periods, movement towards the WE and deposition on its layer, which results in a shortage of cell activity [164–166]. Therefore, to lower costs, successful results have been reported when manufacturing CE using other metals (Ni, Pb, and Cu) [167]. Because of its low resistivity, iridium (Ir) exhibits high electrocatalytic activity. Mokurala et al. [168] prepared a CE containing Ir films on FTO substrates using sputtering radio frequency with a PCE of 7.2%. Silver compounds can also be utilized as CE in DSSCs due to their good thermal conductivity and reflectivity with other metals [169]– [170].

CE including alloy Now, Pt has been considered the ideal choice for application as a CE in DSSCs, due to the many challenges it faces; the researchers seek to prepare promising electrocatalysts from Pt alloys with transition metals (Ni, Fe, Cu, Cr, and Pd) while maintaining catalytic activity. The results showed that Pt alloys have higher activity than pure Pt. Yang and his colleagues [171] produced CE from a Pt ternary alloy (Pt/M/Ni, where M [Co, Pd, and Fe] and examined the synergistic effect on the catalytic activity. They observed an improvement in the electrocatalytic performance of ternary alloys with an efficiency of 8.71%.

CE including carbon material Fortunately, a wide range of alternative materials can be used as CE in DSSCs. Due to their unique qualities as carbon-based materials [172]; they are considered promising alternatives to Pt in DSSCs.

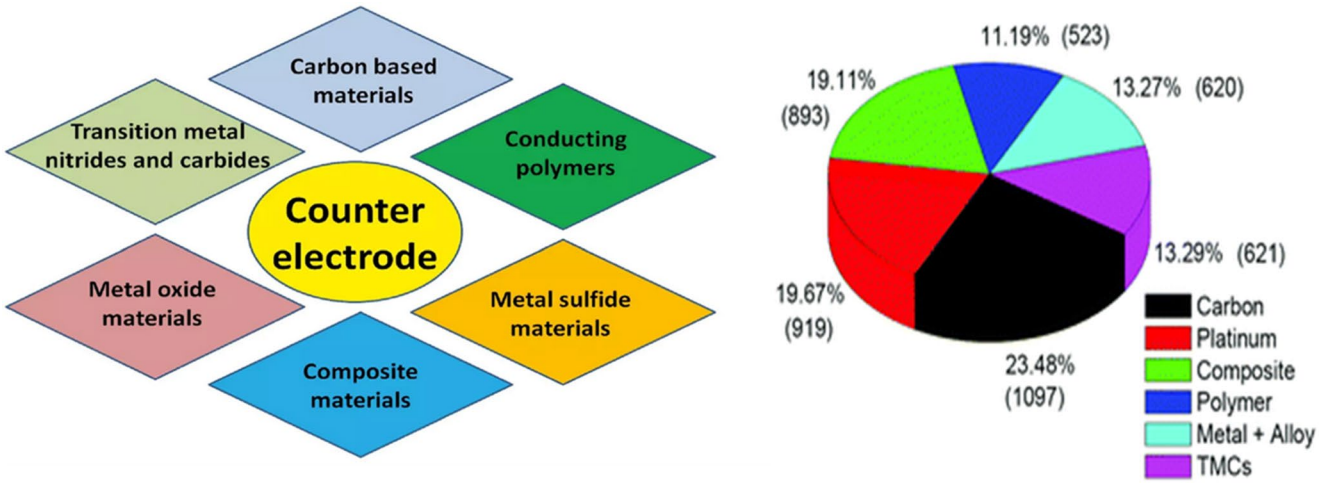

Fig. 11 Different types and percentages of CE applied in DSSCs

A PCE of 14.3% was attained by using an FTO/Au/GNP (graphene nanoplatelets) stack as CE [173]. The competitiveness of carbon materials is increased by their low cost, simple processing, and strong stability. The main drawbacks of conventional carbon-based CEs are that they perform poorer overall than platinized electrodes when combined with the I/I_3 redox pair, both in terms of conductivity and catalytic activity. Additionally, electrode deterioration is caused by poor adhesion to the FTO substrate [174]. In an attempt to address these issues, researchers from Korea University recently doped graphene nanoplatelets with several metals and halogens (I, F, Sb, and Se) to increase the compatibility of carbon CEs towards the I/I_3 redox pair, and these electrodes demonstrated higher stability and efficiency compared to those based on Pt [175, 176].

CE including conductive polymers (CPs) CPs are derivatives of polyacetylene (PE), polyaniline (PANI), polypyrrole (PPy), or polythiophene. These polymers have conductive conjugated double bonds in their molecular structure. CPs exhibit unique light-absorbing properties and can also be utilized as hole-transport materials, making them suitable as charge-transporting materials [177, 178]. Their easy synthesis, excellent conductivity, affordability, abundance, and efficient catalysis in applications in SCs.

CE including inorganic compound Advanced research indicates that inorganic compounds carbon [179], chalcogenides [180], oxygen [181], and nitrogen [182] possess effective catalytic activity, which suggests their utilization as electrodes in SCs. Inorganic compounds have distinctive characteristics, such as a wide range of materials, high flexibility, and ease of manufacture [183], compared to carbon materials and organic polymers, which are enhanced for application as CE in DSSCs.

$DSSC_S$ for climate change solutions

One of the most important actions to confront climate change is to reduce reliance on fossil fuels for energy production, as they are the primary source of greenhouse gases (GHGs), Fig. 12. This can be achieved through alternative options, such as renewable energy (Solar, wind, hydroelectric, geothermal, wave, and tidal energy), while rationalizing operations and enhancing the efficiency of renewable energy. DSSCs, in particular, are an ideal choice for powering autonomous (IoT) devices. This new generation of technology is an intelligent combination of automation and computational power that provides sensing, data processing, artificial intelligence, and ad networking capabilities. (IoT) aims to make human life easier, more pleasant, and more automated by combining efficiency and productivity. IoT devices require an uninterrupted power supply that can run efficiently, and DSSCs have gained popularity due to their exceptional ability to power actuators, sensors, and other IoT communication devices [184]. DSSCs are noted for their strong performance in indoor/ambient light environments, with the potential to remove hurdles to the successful creation of "independent," "smart," and "portable" devices. DSSCs are excellent for portable devices powered by indoor illumination, including incandescent, LED, and fluorescent lights. DSSCs are widely used to power sensors in healthcare, industrial automation, and environmental monitoring. In interior locations with low natural light, DSSCs have been incorporated with building automation systems and electrical appliances to lessen dependency on

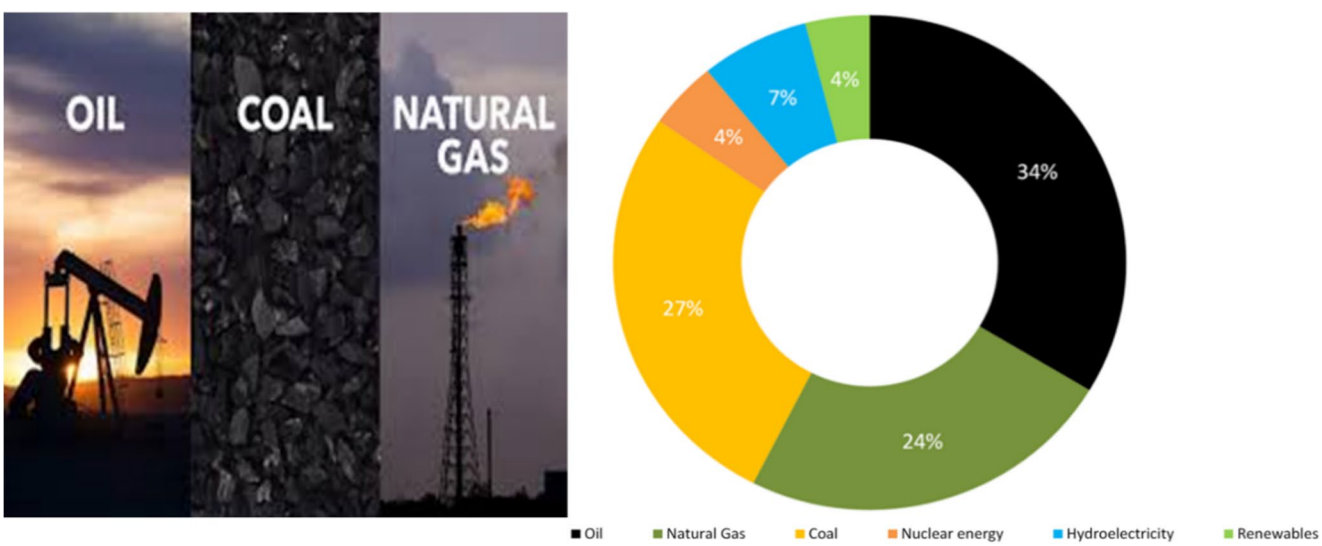

Fig. 12 Consumption curve of Fossil Fuels 2022

finite fossil energy sources. DSSCs can be utilized in photodynamic therapy (PDT) to limit malignant cell proliferation. These PV devices emit light that activates photosensitizing drugs, creating oxygen, which is known to destroy cancer cells. Biomedical implants, such as medication delivery systems or pacemakers, also rely on DSSCs for power. The capacity of DSSCs to generate energy from inside light outperforms that of traditional silicon solar cells. As a result, DSSCs have been deployed as a power supplier in areas where diffuse light predominates over direct solar light. They have been fitted into building windows to ensure that the structures generate their own energy. Transparent and coloured DSSC sheets have also been built up well on glass facades, common windows, and skylights to convert them into power producers. G24 Power, for example, offers Bluetooth-enabled wireless keyboards [185].

In addition, SCs can be applied in water desalination systems and thermal energy generation, depending on other energy sources that stimulate and modify the climate. SCs have advantages that make them the ideal solution to help mitigate climate change. **Low Environmental Impact**: The manufacturing process of DSSCs involves fewer toxic materials compared to traditional silicon-based SCs. Additionally; the use of abundant and non-toxic materials such as titanium dioxide and organic dyes makes DSSCs environmentally friendly. **Cost-Effectiveness**: DSSCs have the potential to be cost-effective due to their simpler and less expensive manufacturing processes. The use of low-cost materials, such as titanium dioxide nanoparticles, and the ability to fabricate them on flexible substrates can contribute to reducing the overall cost of solar energy production. **Efficiency in Low-Light Conditions**: DSSCs exhibit good performance under diffuse or low-light conditions, making them suitable for areas with less sunlight or during cloudy weather. This characteristic makes DSSCs a viable option for solar energy generation in various geographical locations. DSSCs have shown promising efficiency levels, with the current record exceeding 15.2%. While this is lower than the efficiency of silicon-based SCs (which can exceed 20%), DSSCs have other advantages that make them attractive, such as their ability to perform well under low-light conditions and their potential for cost-effective manufacturing. One of the most important features of SCs is their **long-term stability.** Practically, the system composed of **FTO/WS_2/N719/CBTS/Au** has demonstrated stability for 90 days [44]. However, further research on long-term stability and efficiency is necessary to develop commercially competitive products. For indoor applications, a lifetime of at least 5 years is required, while for outdoor applications, a lifetime exceeding 15 years is needed to ensure competitiveness [185]. **Architectural Versatility**: DSSCs can be fabricated in different shapes, sizes, and colours, allowing for architectural integration and design flexibility. They can be incorporated into building facades, windows, or other surfaces, enabling the generation of solar energy without compromising aesthetics. In general, energy devices derived from solar energy are considered more environmentally friendly and a promising solution to climate change, unlike energy derived from fossil fuels, which has been the most widely used energy source over the past decade. It has been allocated 84.2% of total energy consumption in 2022, and its completion rate has reached approximately 1.2 million terawatt-hours at present (Fig. 12). However, there are two major disadvantages to fossil fuels. First, they are non-renewable fuels, whether on the scale used on Earth, and the climate changes and environmental damage they cause.

In just a few years, the burning and production of fossil fuels for energy and transportation have released approximately 120 million metric tons of methane and billions of tons of carbon dioxide into the atmosphere annually. As shown in Fig. 13, Fossil fuel emissions trap heat in the Earth's atmosphere, causing climate change have increased

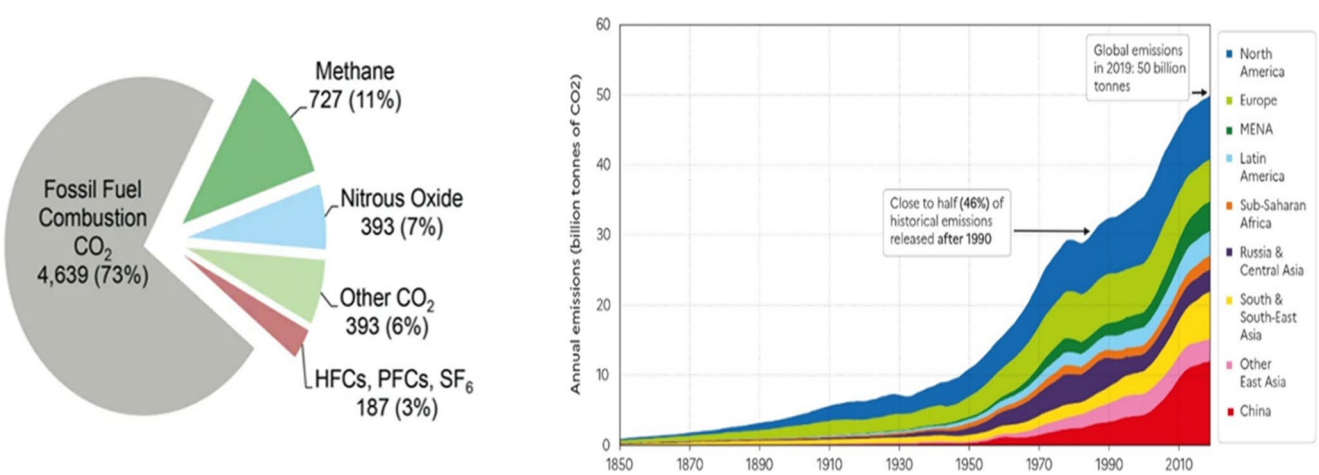

Fig. 13 U.S. GHG emissions, 2021

dramatically over the previous 70 years, reaching 50 billion metric tons in 2019 compared to barely 5 billion metric tons in 1950.

The problems causing climate change as a result of gases emitted as a result of the combustion of fossil fuels are many and varied (Fig. 14), the most important of which are:

Ocean acidification: The Ocean absorbs at least 25% of the carbon dioxide released by burning fossil fuels, altering its chemistry (pH). It is more difficult for marine species to form coral skeletons and shells due to the increased acidity. Ocean acidity has revolt 30% in the last 150 years, endangering fishing, coral reefs, tourism, and the economy. **Degradation of land and water**: landscapes and ecosystems suffer greatly from the extraction, processing, and transportation of subterranean coal, gas, and oil resources. Large areas of land are leased by the fossil fuel sector for processing, waste storage, and disposal facilities, as well as infrastructure, including pipelines, access roads, and wells. When strip mining occurs, large areas of land, including forests and entire mountaintops, are blasted and scraped to reveal subterranean coal or oil. The nutrient-leached land will never recover to its previous state, even when operations end. This leads to the fragmentation and destruction of vital wildlife habitat, which is essential for breeding and migration. Even animals that can leave can end suffering because they are frequently compelled to live in less-than-ideal habitats and have to compete violently with native wildlife for limited resources. The decline in water levels is also considered one of the most serious dangers of climate change; population increase, urbanization, climatic anomalies, and global warming all contribute to chronic water scarcity. For example, Africa is home to the world's longest river, the Nile, as well as several other large rivers, including the Congo, Niger and Zambezi. At the same time, Africans have significant challenges in obtaining safe drinking water. More than 40% of the world's population suffers from water shortage, which is expected to worsen in the next decades. The urban population in emerging countries is expected to grow rapidly, resulting in increased water demand. Africa is the second-largest continent with a population of 1.3 billion people, resulting in steady growth in the consumption of commercial products, water, and energy. Energy data suggest that just one in every four individuals in Sub-Saharan Africa has access to electricity. Long-term energy demand and supply projections are crucial in Africa due to the ongoing rise in energy consumption and the scarcity of adequate financial resources [186, 187].

Fig. 14 Climate change impacts

Extreme weather: The National Oceanic and Atmospheric Administration claims that the use of fossil fuels contributes to climate change, which in turn causes extreme weather events to occur more frequently and with greater severity, leading to disasters that cost at least $1 billion. The US spent an estimated $606.9 billion between 2016 and 2020 in response to extreme weather events, including hurricanes, storms, flooding, and droughts. **Sea level rise**: Glaciers and land ice sheets are melting due to rising ocean and atmospheric temperatures, which are caused by climate change. Since the late 19th century, sea levels have risen about 9 inches. As a result, there have been more frequent floods, powerful storms, and saltwater intrusions. It is projected that protecting coastal areas from sea level rise could cost $400 billion over the next 20 years, with 40% of the country's population residing near the coast. **Pollution of Air**: Air pollution: the combustion of fossil fuels releases several hazardous air pollutants that harm the environment and human health: sulfur dioxide, nitrogen dioxide, carbon monoxide, and mercury. Acid rain, eutrophication (an excess of nutrients that lowers oxygen levels in aquatic environments), damage to crops and forests, and harm to wildlife are all consequences of air pollution. **Water pollution** is caused by fossil fuels, which can be found in fracking fluids and oil spills. A fracking well can require 1.5 million to 16 million gallons of water. The wastewater that is left behind can be hazardous and frequently contains elements that can contaminate drinking water and groundwater, such as arsenic, lead, and mercury.

The future vision of reducing climate change using solar energy

The International Renewable Energy Agency (IRENA) is a global hub, an important arena for global cooperation, and a meeting place for experts in finance, technology, resources, and policy specialization in renewable energy. To achieve sustainable development, promote energy security, drive low-carbon economic growth, and achieve a prosperous society, the agency has been working to adopt and use all forms of renewable energy on a large and sustainable scale, including thermal, hydropower, ocean, solar, and wind energy. This work has been done since the agency was established as a government organization in 2011. For example, in the United States, GHGs emissions are typically produced by a wide range of economic sectors, the largest contributors to which are the transportation and electric power industries (28%), industry and commercial sector (22%). Figure 15 show residential consumption (11%) and agriculture (9%). The country's attempts to cut emissions and meet stringent climate goals rely heavily on solar technology, and the most important option here is to expand the use of solar energy in a variety of environmental fields.

At the Electric Sector, the PV and concentrated solar power (CSP) systems may both provide clean, emission-free electricity that can be fed directly into the city grid. It is also possible using solar heating and cooling (SHC) systems. As of the first half of 2023, the aggregate installed solar electricity capacity in the United States was over 155.2 gigawatts, which is sufficient to power over 27 million ordinary American households.

Transport sector Electric: Vehicles and hybrid vehicles are widely viewed as a near-term climate change solution in the transportation sector, especially when these vehicles are charged by a solar garage or charging station. As distributed solar systems become more widespread, electric vehicles play a bigger role in lowering global carbon emissions by creating a comprehensive system that includes rooftop solar, home battery storage, and electric vehicles.

Sectors: Residential and Commercial. While residences and apartments make up the residential sector, structures

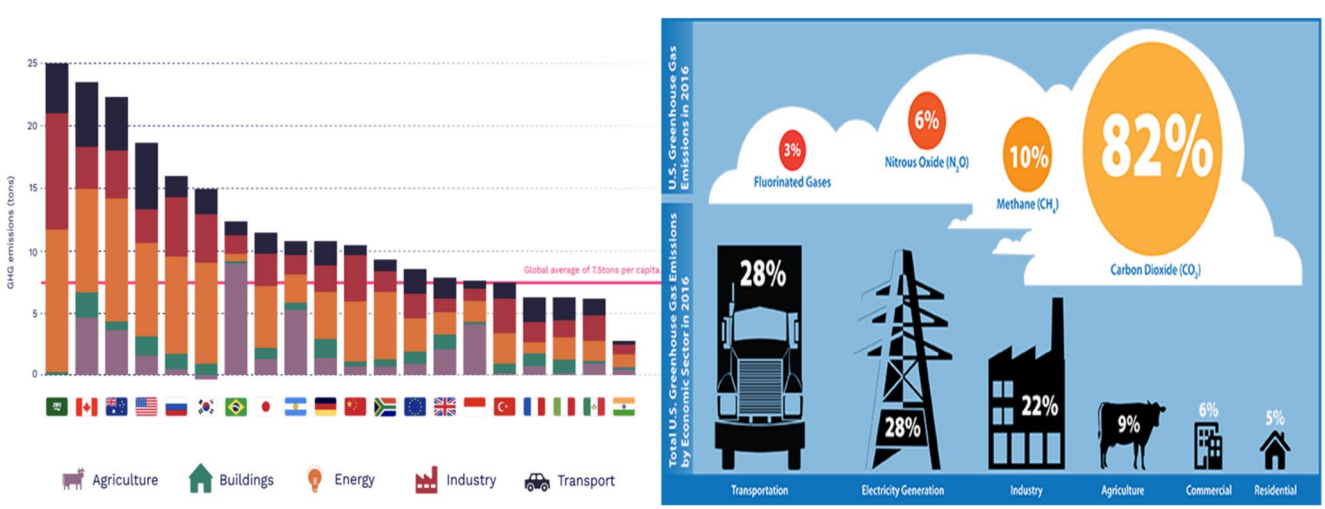

Fig. 15 CO_2 emission by different country and sectors

such as offices, malls, warehouses, schools, restaurants, and hospitals are included in the commercial sector. The majority of energy used in buildings, whether residential and commercial, is used for water heating, space cooling, and space heating. Given that SHC systems can supply over 80% of the energy required for space heating and water heating requirements, this is an ideal use case for SHC technology. In addition, solar air conditioning offers an environmentally friendly and emission-free alternative to energy for satisfying cooling requirements. Many of the biggest companies and enterprises prefer to employ solar PV to generate electricity for their warehouses, headquarters, and distribution centers. These companies use massive rooftop systems or off-site solar power. Although the greenhouse gas (GHG) emissions during the life cycle of solar PV cells are up to 20–70 g CO_2e/kWh, depending on the system age and specific manufacturing details, compared to silicon PV cells that show lower emissions in the range of 13–30 g CO_2e/kWh, it remains a promising energy alternative. In an environmental comparison of electricity generation from DSSC system to a natural gas/combined cycle power station, the gas power plant would produce 450 g CO_2/kWh, whereas the DSSC system would produce 19–47 g CO_2/kWh. The latter is comparable to 42 g CO_2/kWh for an amorphous silicon-based solar cell system [188–190]. The US Energy Information Administration recently revealed that carbon dioxide emissions in the US have dropped by 28% since 2005 as a result of the widespread use of solar energy. Consequently, adding solar panels to your home is a highly efficient way to lessen your carbon footprint Fig. 16. Decades of clean energy generation, water conservation, and energy cost savings are left behind when solar panels take two to three years to offset their life cycle emissions.

Future vision of solar energy for climate changes solutions in Egypt

The Egyptian state has developed a road map through which it will proceed on parallel paths to deal with these challenges, especially with regard to the expansion of green economy projects and the development and diversification of renewable energy sources and maximizing their value, as it is one of the safe havens to undermine the causes of climate change and reduce the economic burden resulting from the uses of traditional energy sources. Egypt is striving to unite global efforts at the climate summit (COP 27) in Sharm El Sheikh in order to formulate a joint vision to confront climate change and its negative impacts, in addition to growing partnerships with the private sector and international partners in the fields of clean and renewable energy and working to provide the necessary technology in this field, making it eligible to become one of the largest producers of clean energy. The government seeks to expand reliance on renewable energy, explaining that the sustainable energy strategy for 2035 aims to increase the contribution of the percentage of renewable energy to the electrical energy mix, as electricity production from renewable energy is scheduled to reach 42% of the total electrical energy produced in 2035, compared to 20% in 2022. The Egyptian government has paid special attention to solar energy due to Egypt's distinguished location, as the capacity of the implemented solar energy projects reached 1,756 megawatts, the most prominent of which are the PV cell projects in Benban, 1,465 megawatts, the Karimat Shams Thermal, 140 megawatts, and the PV cells on rooftops, 121 megawatts. In addition, the government reports that there are PV cell projects under study with a capacity of up to 770 megawatts.

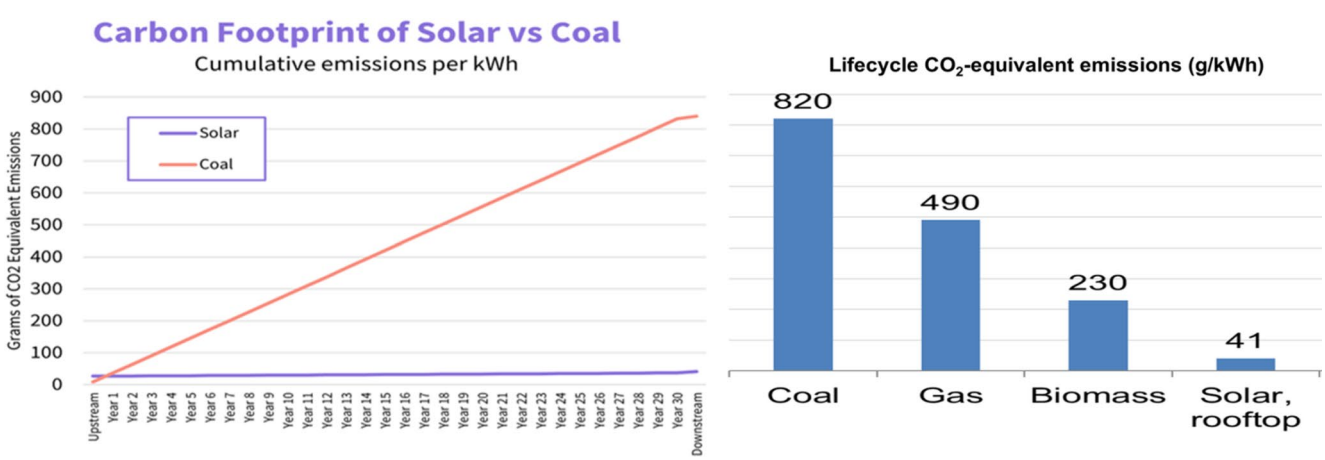

Fig. 16 CO_2 emissions by Solar Energy

Conclusion and future perspective

The problem of climate change is now at the forefront all over the world. The effects are being felt around the world as a result of longer-lasting heat waves, depletion of freshwater reserves, melting ice caps, and irreparable damage to many of the world's ecosystems. If rapid measures are not taken to reduce emissions, the planet will remain trapped in increasing temperatures, and its effects will become more severe and dangerous, threatening many of the world's peoples and natural systems. Most of the world's governments are aware of the existential threat posed by climate change, but their ambition at the present time is to mitigate the changes is not enough. If all countries followed through on the climate goals pledged by the Paris Agreement, this would limit the rise in average global temperature to 2.6–3.2 °C. Recently, governments have begun to move seriously to solve the climate change crisis by the circular economy and alternative energy sources. The circular economy has received significant attention as a promising approach to addressing the challenges of resource depletion, environmental degradation, and unsustainable consumption patterns. This study examines the application of the circular economy framework to Sustainable Development Goal 6 (which focuses on ensure availability and sustainable management of clean water and sanitation for all). From a circular economy perspective, it explores how to integrate the principles of reduce, reuse, recycle, and recover into water management, wastewater treatment, and sanitation systems to achieve this goal. Through case studies, policy papers, and innovative practices from around the world, it identifies synergies between the circular economy and SDG 6 targets, such as reducing water waste, improving water efficiency, and recovering valuable resources from wastewater [191]. Renewable energy is considered a promising solution to get out of this crisis, and there are different types, but solar energy has priority in generating electricity due to its many characteristics that were mentioned previously. We can see the future with solar energy in many sectors of agriculture in some developing countries, the cost of growing crops is very high due to the cost of electricity, as the agricultural parts needed to pump water from rivers or wells require a lot of energy. Solar-powered pumps can be used to solve this crisis, as enjoying the advantages of solar pumps allows them to work for a longer time, and adding batteries or other storage systems enables us to obtain additional energy generated by solar energy at night instead of relying on it only during the day. We can also benefit from groundwater by extracting it using solar energy pumps and storing it until it can be used for agriculture and irrigation, especially in the desert environment, which helps ensure the presence of permanent and continuous energy for the agricultural and food sector, ensure food production for consumers, and prevent a sudden rise in prices through which it can be created. A state of economic imbalance, also improving food security: enhancing food security by reducing production costs, as it is possible to use solar energy for drip or spray irrigation, especially for crops that require a lot of water, which reduces farmers' dependence on water, increases production efficiency, produces agricultural crops at a low cost, increases profits, and provides them with food security. Solar panels can also be relied upon to cool and dry vegetables and fruits, and this will help preserve crops for long periods. Solar water heating can provide hot water for livestock, warm and stimulate grazing cows, and use solar energy for electricity to light agricultural farms. Also, for harvesting crops and plowing, cars can be used for plowing purposes that are powered by solar energy. These SCs save electricity and also eliminate pollution and climate changes. In industry, solar energy networks can be applied in factories where factories need to generate the high heat necessary to carry out their series of tasks without harmful emissions, including metal processing. Chemical products, food products, water desalination as well. Mining Solar panels can be installed at mining sites to capture sunlight and convert it into electricity. This sustainable energy source can be used to power various mining equipment, including drills, crushers and conveyors. Excess energy can be stored in batteries or fed back to the grid, making mining operations more energy efficient and environmentally friendly. They are also used in the field of medicine and healthcare in operating medical devices.

Author contributions Samar H. Bendary: Conceptualization, Methodology, Investigation, Validation, Writing original draft, writing review & editing. Sawsan A. Mahmoud: Conceptualization, Methodology, Investigation, Data curtain and Supervision.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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